



AKTU

B.Tech Sem-6

CS IT & CS Allied

CN: Computer Networks

ONE SHOT Revision

Unit-5 :Application Layer

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- M.Tech Gold Medalist
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Computer Networks

Unit-5

AKTU Syllabus

Application Layer: ^{Imp} Domain Name System, World Wide Web and ^{HTTP} Hyper Text Transfer Protocol, ^{SMTP, POP3, IMAP Imp} Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography – basic concepts.

↓
TELNET

↓
SNMP

↓
RSA algo

AKTU PYQs

1. Explain DNS. (AKTU 2022-23, 2023-24, 2021-22, 2018-19, 2017-18, 2016-17)
2. Explain HTTP and its uses. (AKTU 2021-22, 2017-18)
3. State difference between HTTP and HTTPS (AKTU 2022-23)
4. "Elaborate about TELNET and its working procedure." (AKTU 2016-17, 2017-18, 2018-19, 2023-24)
5. What is the purpose of the Domain Name System? Discuss the three main divisions of the domain name space. (AKTU 2018-19)
6. Explain SNMP in detail. (AKTU 2023-24)
7. Differentiate between FTP and HTTP. (AKTU 2023-24)
8. Explain the following terms:
9. (i) FTP (ii) SMTP (AKTU 2022-23, 2021-22)
10. Define SNMP Protocols and working scenario (AKTU 2022-23)

...AKTU PYQs

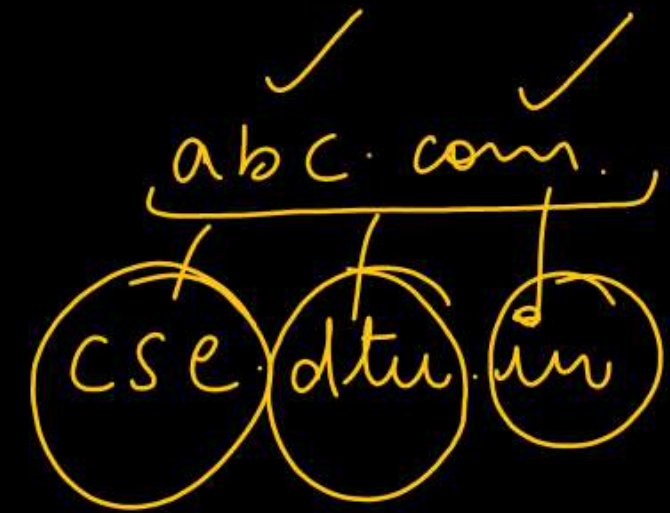
11. Define Compression. (AKTU 2023-24)
12. Explain Cryptography. Write about RSA Algorithm with example. (AKTU 2023-24)
13. Explain Asymmetric cryptography. Also, write the steps used in RSA algorithm, demonstrate the transmission of character "F" using RSA. (AKTU 2022-23)
14. List out few email gateways. (AKTU 2016-17)
15. (i) How is TFTP different from FTP?
(ii) What three functions can SNMP perform to manage network devices? (AKTU 2018-19)
16. What is the difference between a user agent (UA) and a mail transfer agent (MTA)? (AKTU 2018-19)

Domain Name System (DNS)

- DNS translates human-readable domain names (e.g., www.google.com) into machine-readable IP addresses (e.g., 142.250.190.14), enabling internet communication.
- It enables computers to locate and communicate with each other on the internet.
- Functions as a hierarchical, distributed database.

- Queries pass through multiple levels:

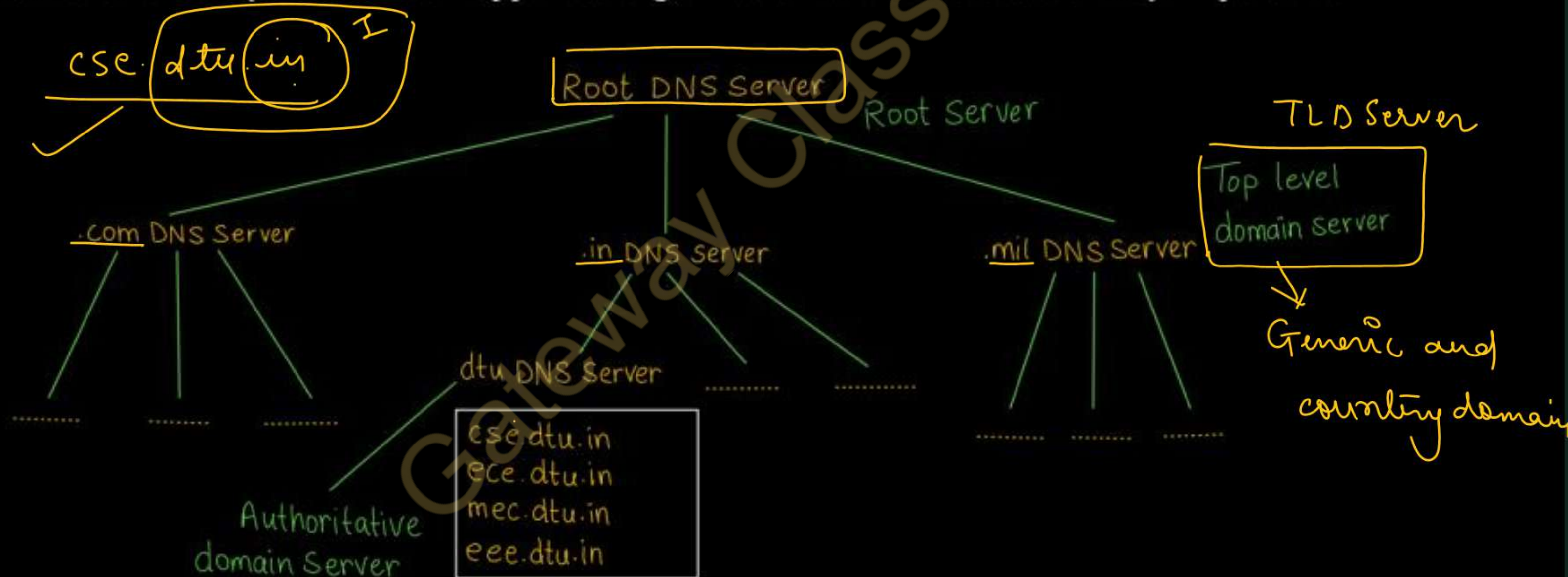
- Root server (13)
- Top-Level Domain (TLD) server
- Authoritative server (stores the specific IP address).



- Ensures seamless website access using easy-to-remember names instead of numerical IP addresses.

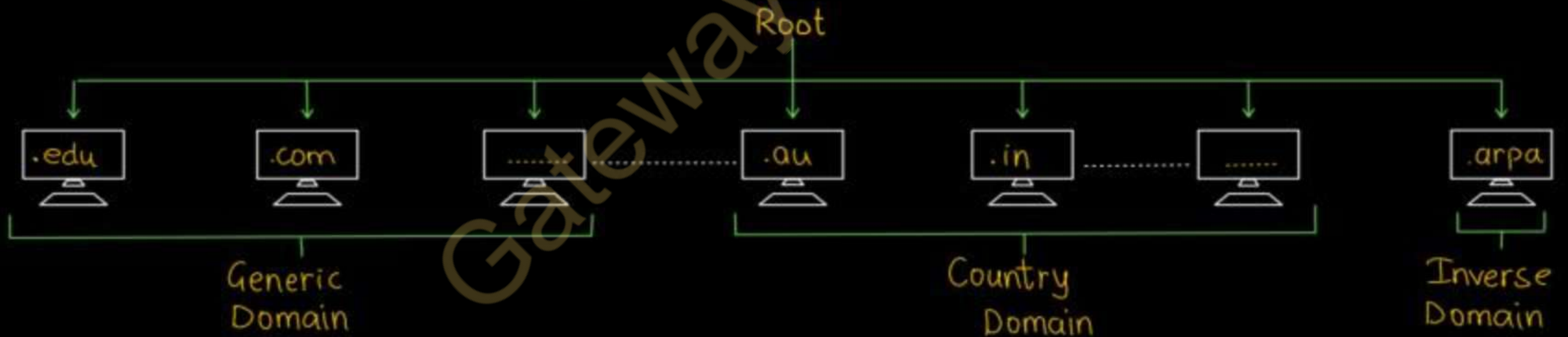
Structure of DNS

- It is very difficult to find out the IP address associated with a website as there are millions of websites and we should be able to generate the IP address for all of them immediately, there should not be a lot of delays for that to happen so organization of the database is very important.



Types of Domain

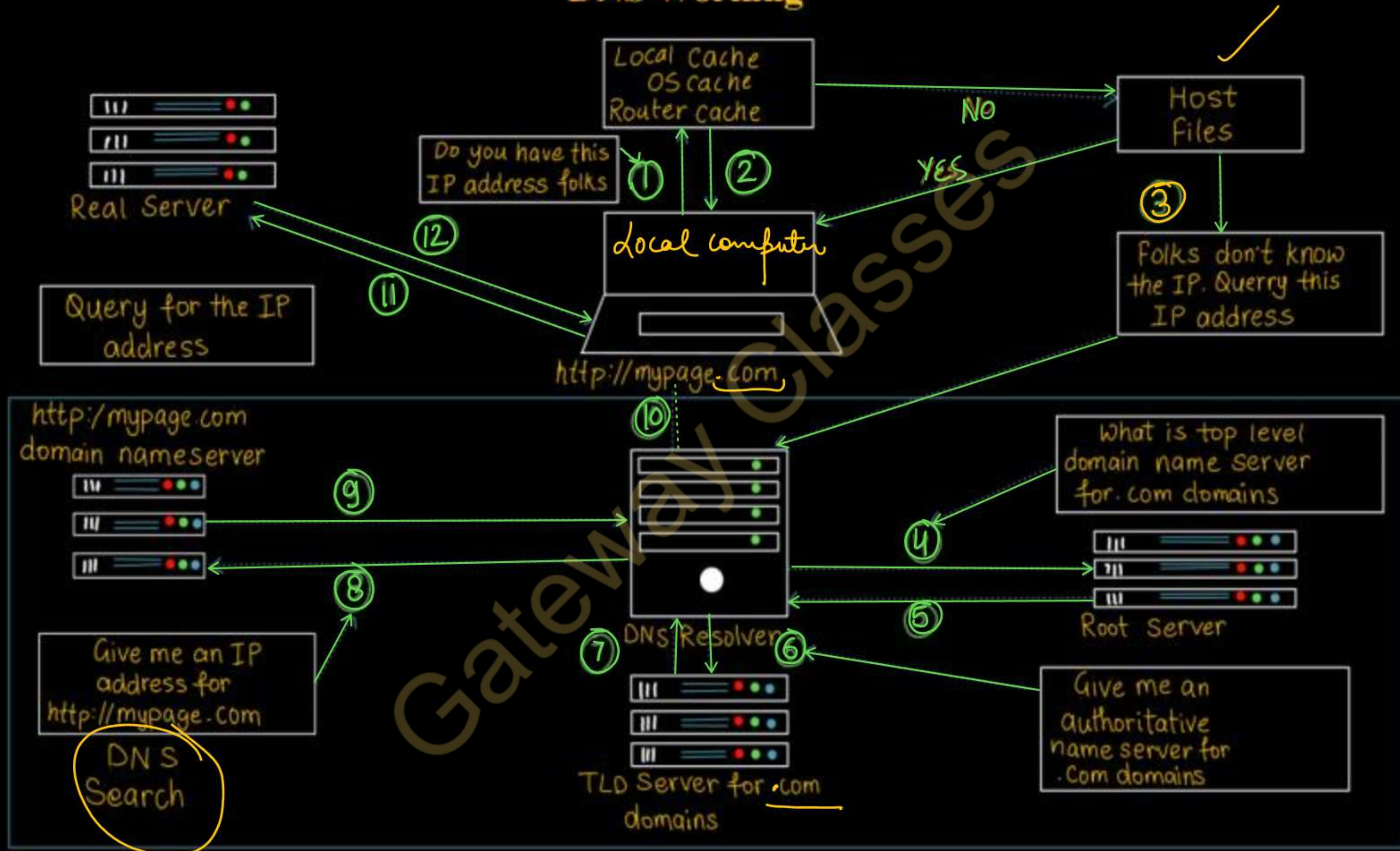
- **Generic Domains:** .com(commercial), .edu(educational), .mil(military), .org(nonprofit organization), .net(similar to commercial). All these are generic domains.
- **Country Domain:** .in (India) .us .uk
- **Inverse Domain:** if we want to know what is the domain name of the website. IP to domain name mapping. So, DNS can provide both the mapping.
- To find IP addresses of **www.hello.org**, we have to type **nslookup www.hello.org**



DNS Working

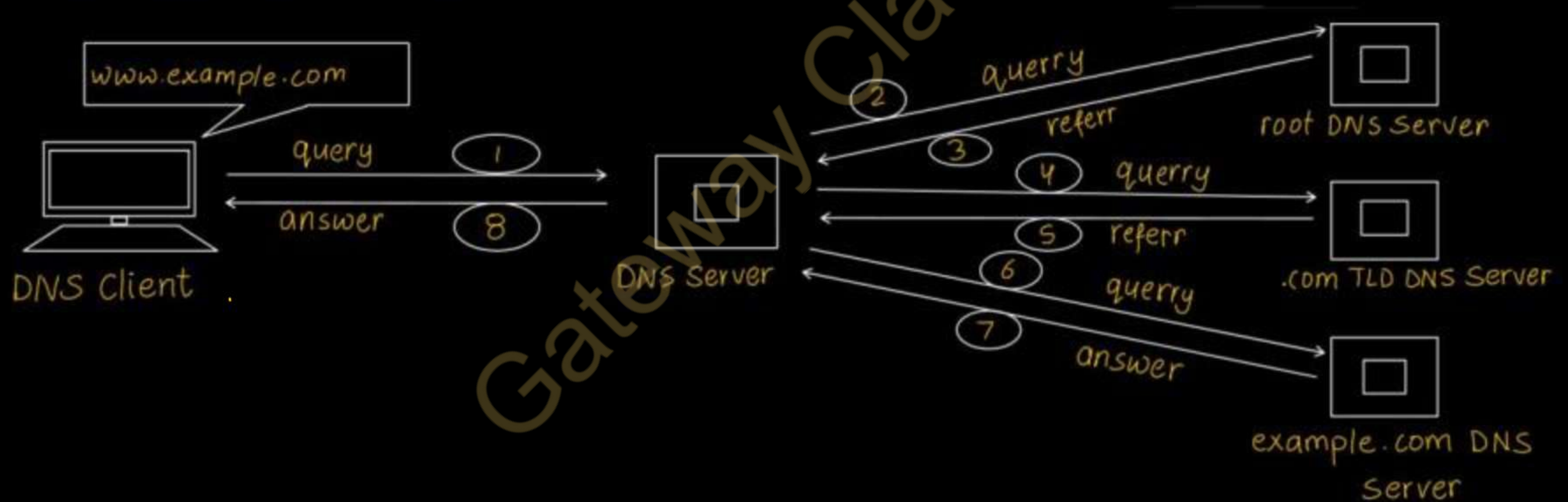
- We type a website like <https://mypage.com> in our browser, our computer tries to find the IP address.
- First, it checks local cache (our browser, OS, or router) to see if it already knows the IP address.
- If local cache doesn't have IP, check host files (used for specific manual mappings).
- If still not found, the query is sent to a **DNS resolver** (usually provided by your ISP).
- DNS resolver sends query to a **Root DNS server**, which points to Top-level domain (TLD) server (.com server for this example).
- **TLD server** then directs the resolver to the authoritative name server for **mypage.com**.
- **Authoritative nameserver** knows the exact IP address for **mypage.com** and sends it back to the resolver. This server holds the actual DNS records and responds with the IP address.
- Resolver passes the IP address to our computer.
- Our computer uses the IP address to connect to the real server where the website is hosted.
- The website loads in our browser.

DNS Working



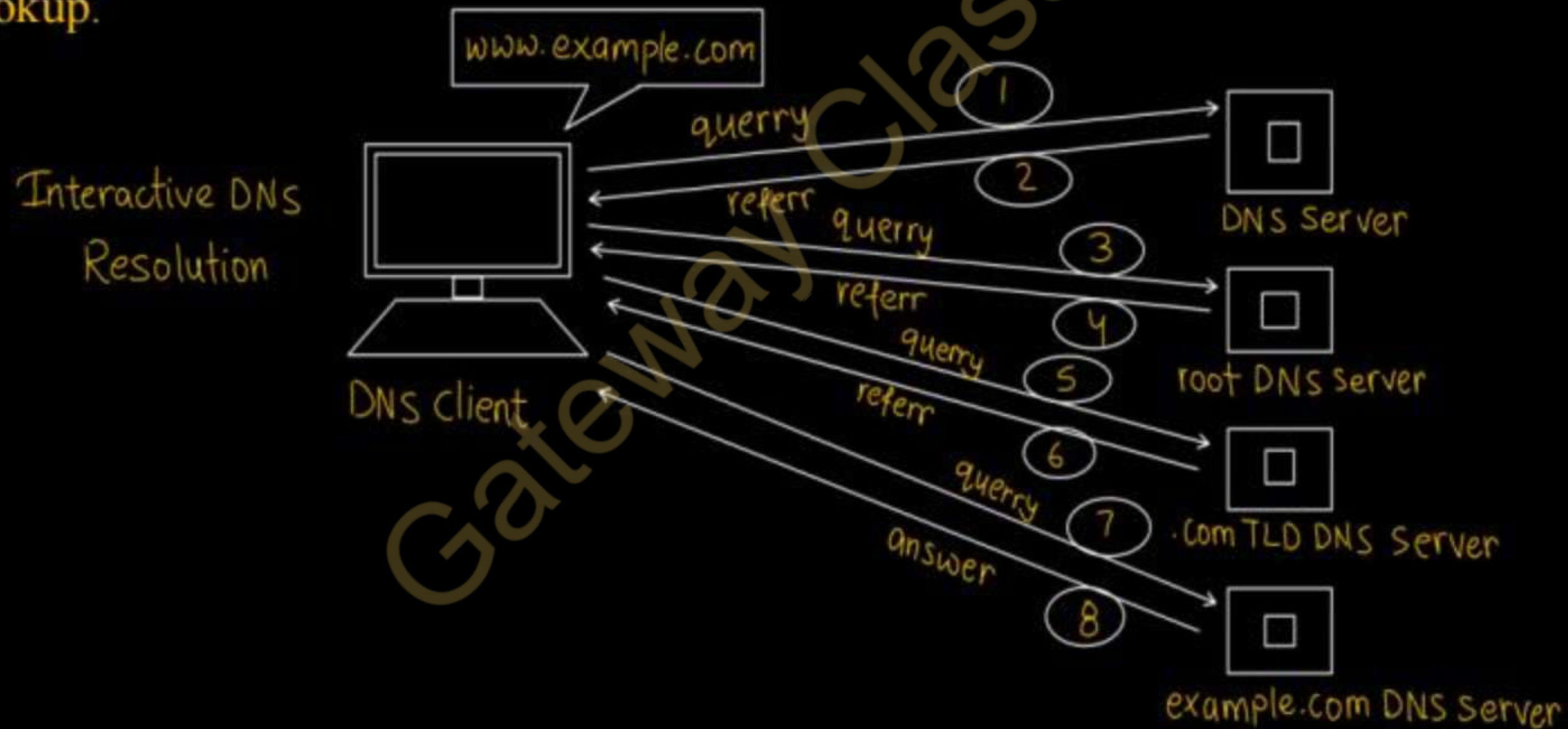
DNS recursive query:

- DNS server, who received your query will do all job of fetching the answer, and giving it back to you.
- During this process, the DNS server might also query other DNS servers in the internet on your behalf, for the answer. **One DNS server communicates with several other DNS servers to find an IP address and return it to the client.**



Iterative DNS query:

- In this, the DNS server, will not go and fetch the complete answer for your query, but will give back a referral to other DNS server's, which might have the answer.
- In an iterative **DNS query**, where the client communicates directly with each **DNS** server involved in the lookup.



Caching:

- When a server asks for a mapping from another server and receives the response, it stores this information in its cache memory before sending it to the client.
- If the same or another client asks for the same mapping, it can check its cache memory(server marks the response as *unauthoritative*)and solve the problem.

Problems:

If a server caches a mapping for a long time, it may send an outdated mapping to the client.

To counter this, the authoritative server adds TTL to the mapping.

Importance of DNS

➤ **User-Friendly Navigation:**

- DNS simplifies user navigation by allowing users to access websites using easy-to-remember names instead of dealing with complex numerical addresses.

➤ **Load Balancing:**

- Help distribute traffic across multiple servers and ensures efficient load management and enhances website performance during traffic periods.

➤ **Email Routing:**

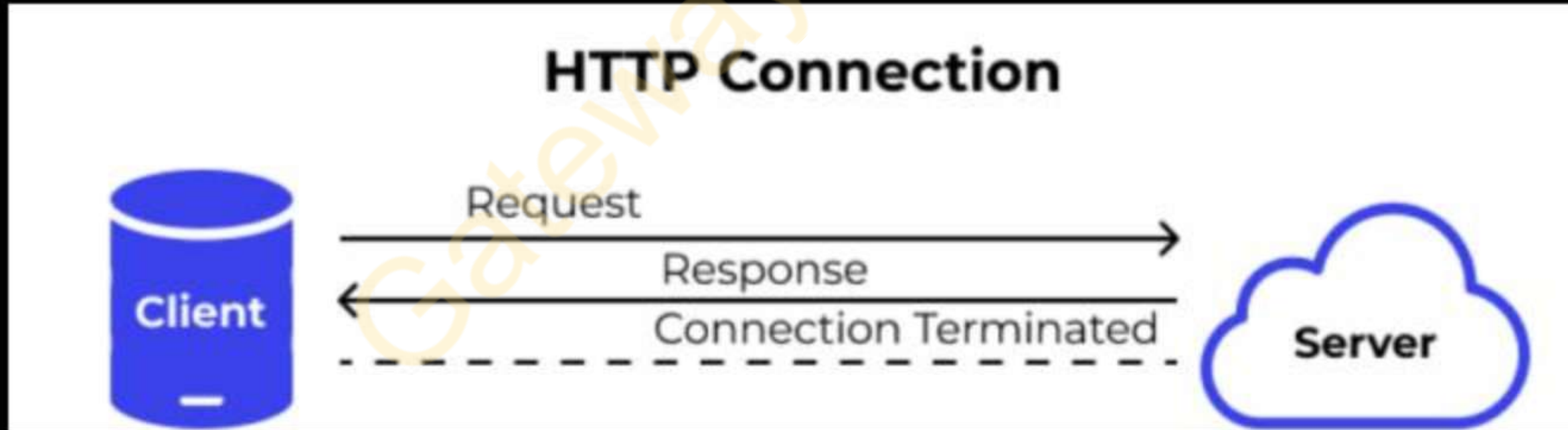
- Enable email routing by directing emails to the correct mail servers associated with a domain.

➤ **Security Features:** adds extra layer of security ensuring DNS responses are authentic not tampered.

➤ **Scalability:** DNS provides scalability by allowing organizations to add subdomains or new services without significant changes to their overall infrastructure.

HTTP (Hypertext Transfer Protocol)

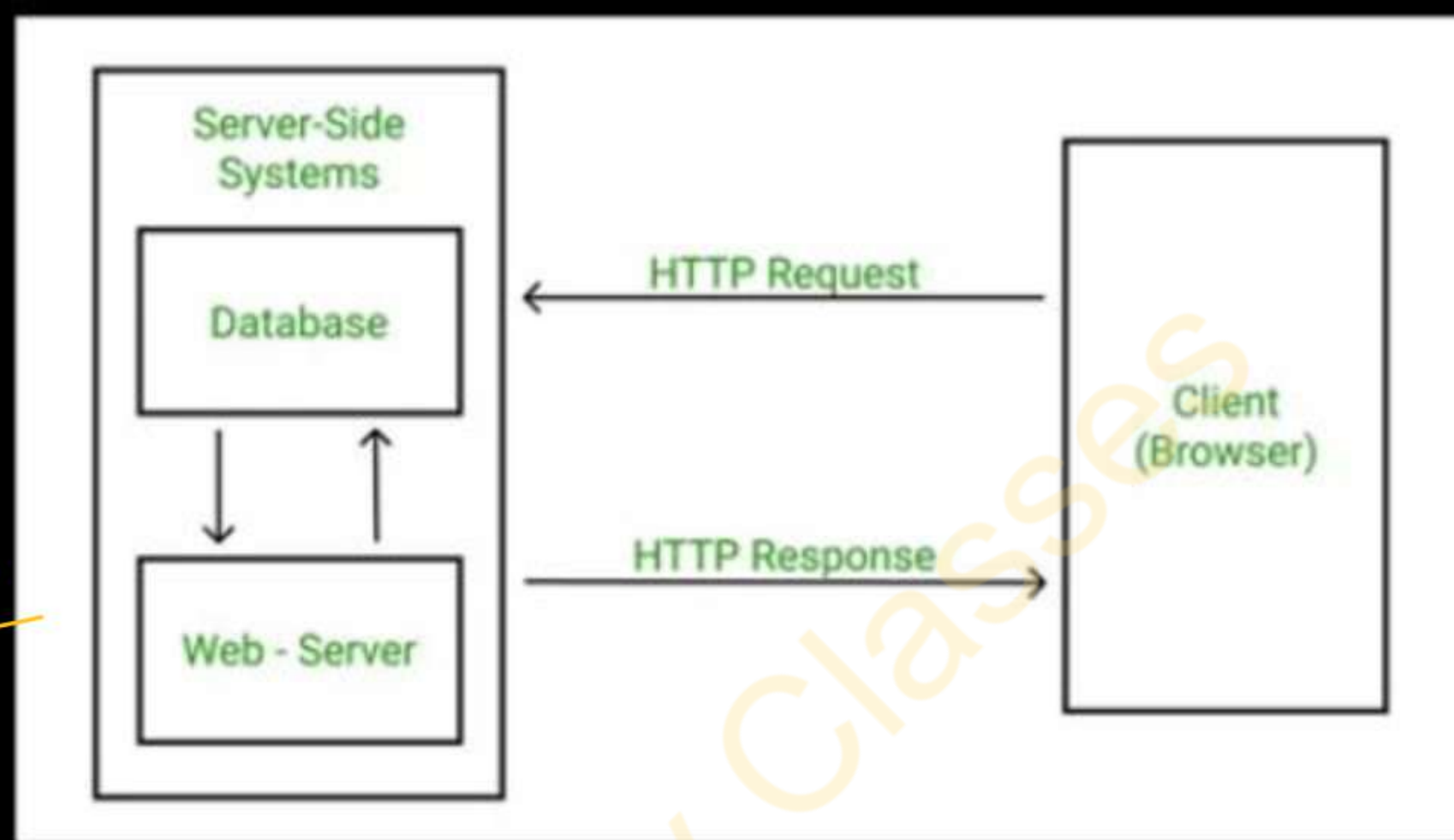
- It is a fundamental protocol of the Internet, enabling data transfer between a client and a server.
- It is the foundation of data communication for the World Wide Web. port no. 80
- HTTP provides a standard between a web browser and a web server to establish communication.
- A set of rules for transferring data (text, images, and multimedia files) from one computer to another.
- Whenever a web user opens their web browser, the user indirectly uses HTTP.



Methods of HTTP

- **GET**: Used to retrieve data from a specified resource.
- **POST**: Used to submit data to be processed by a specified resource.
- **PUT**: Used to update or create a resource on the server.
- **PATCH**: Similar to PUT but used for partial modifications to a resource.
- **DELETE**: Used to remove a specified resource from the server.
- **HEAD**: Similar to GET but retrieves only the response headers, useful for checking resource properties without transferring the full content.
- **OPTIONS**: Used to retrieve the communication options available for a resource, including supported methods and headers.
- **TRACE**: Used for debugging purposes to echo the received request back to the client.
- **CONNECT**: Used to establish a tunnel to the server through an HTTP proxy, commonly used for SSL/TLS connections.

80



Features:

- Stateless: Each request is independent, and the server doesn't retain previous interaction information.
- **Text-Based**: Messages are in plain text, making them readable and debuggable.
- **Client-Server Model**: Follows a client-server architecture for requesting and serving resources.
- **Request-Response**: Operates on a request-response cycle between clients and servers.
- **Request Methods**: Supports methods like GET, POST, PUT, DELETE for different actions.

Difference Between HTTP and HTTPS (AKTU)

HTTP(HyperText Transfer Protocol)	HTTPS(HyperText Transfer Protocol Secure)
URL begins with "http://".	In HTTPs, URL starts with "https://".
uses port number 80 for communication.	uses 443 port number for communication.
unsecure.	HTTPs is considered as secure.
works at Application Layer.	HTTPS works at Transport Layer.
Encryption is absent.	Encryption is present in HTTPS.
faster than HTTPS	slower than HTTP
Does not use data hashtags to secure data.	While HTTPS will have the data before sending it and return it to its original state on the receiver side.
In HTTP Data is transfer in plaintext.	In HTTPS Data transfer in ciphertext.
HTTP Does not require SSL/TLS or Certificates	Requires SSL/TLS implementation with Certificates.

Hello → plaintext
 ↓
 IFMMP → ciphertext

Advantages:

- Platform independence: Works on any operating system
- Compatibility: Compatible with various protocols and technologies
- Efficiency: Optimized for performance
- Security: Supports encryption for secure data transfer

Disadvantages:

- Lack of security: Vulnerable to attacks like man in the middle
- Performance issues: Can be slow for large data transfers
- Statelessness: Requires additional mechanisms for maintaining state

Remote Logging

- Remote Login is a process in which user can login into remote computer and use services that are available on the remote computer. It is implemented using Telnet.

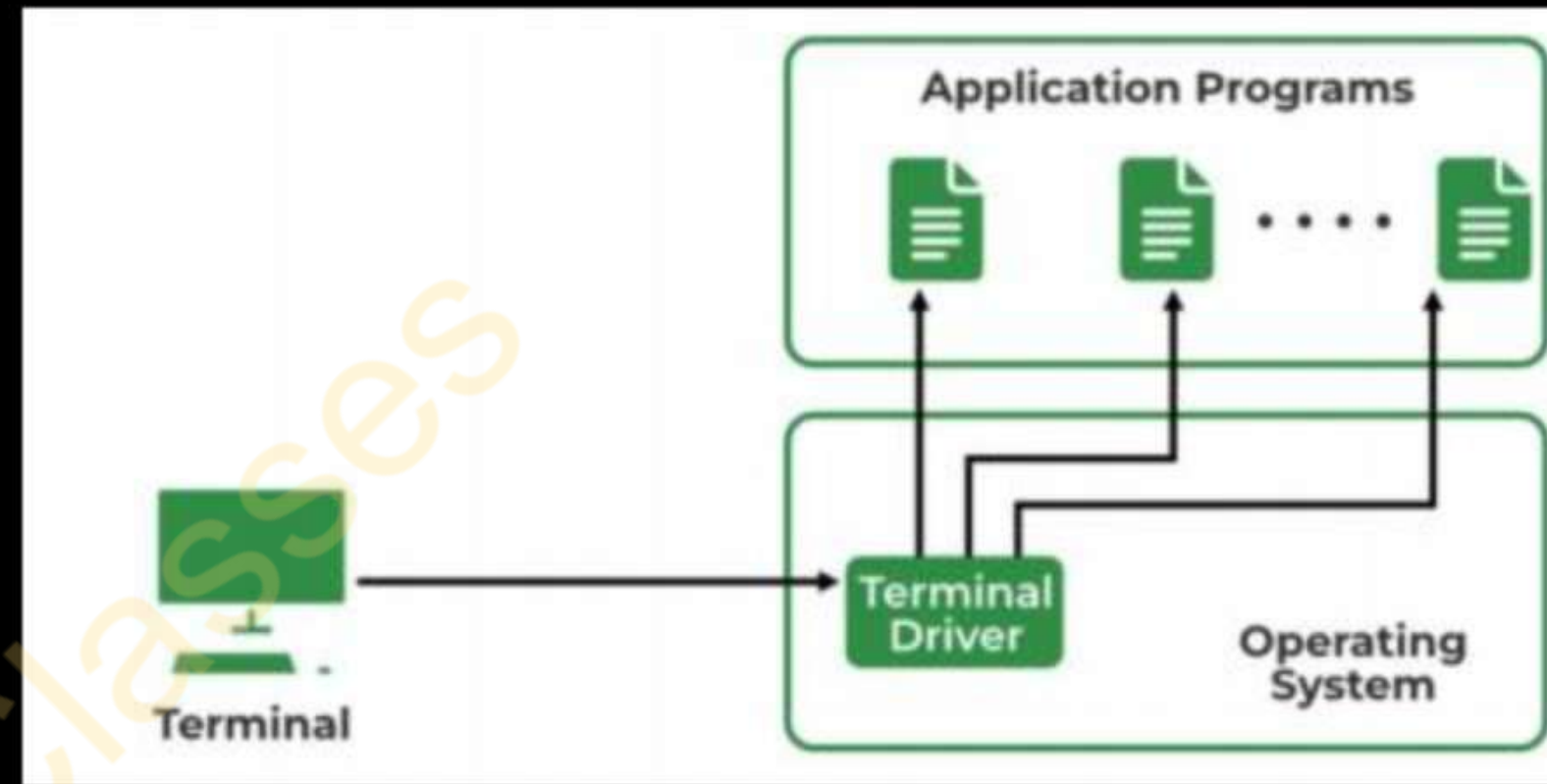
Example:

Students may want to connect to their university computer lab from their home to access application programs for doing homework assignments.

- One way is to create a client/server application program for each desired service using FTP or SMTP but it would be impossible to write a specific client/server program for each demand.
- Better solution is remote login that lets a user access any application program on a remote computer. After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.

Local Login:

- When a user logs into a local timesharing system, it types at terminal, keystrokes are accepted by terminal driver.
- Terminal driver passes the characters to the OS.
- OS interprets the combination of characters and invokes the desired application program or utility.



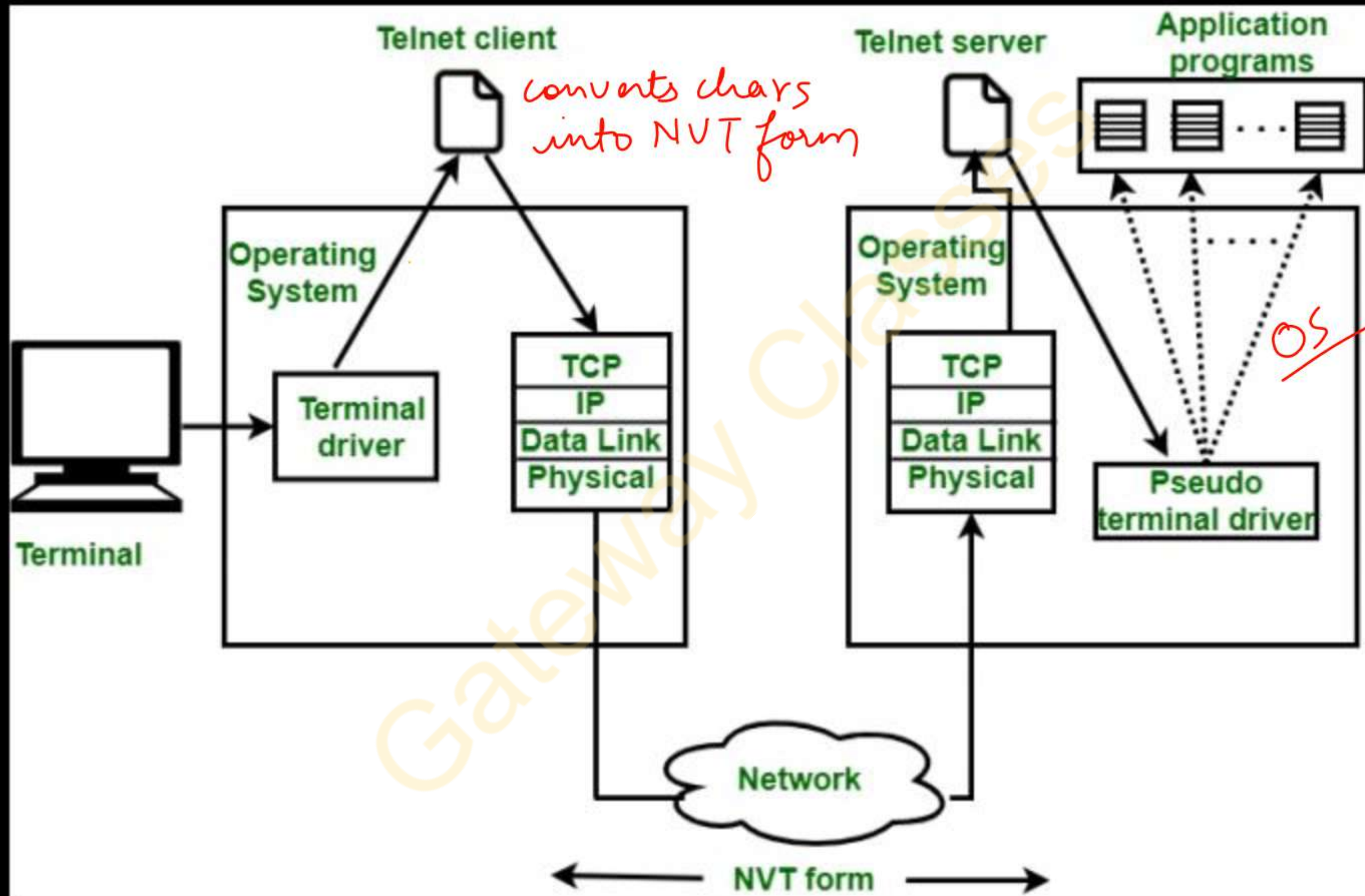
TELNET (TERminaL NETwork):

- It is the standard TCP/IP protocol for virtual terminal service proposed by ISO.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- TELNET is a general-purpose client/server application program.

Procedure of Remote Login :

1. When the user types something on local computer, local OS accepts the characters.
 2. Local computer does not interpret the characters; it will send them to TELNET client.
 3. TELNET client transforms these characters to a universal character set called Network Virtual Terminal (NVT) characters and it will pass them to the local TCP/IP protocol Stack.
-
1. Commands/text in form of NVT, travel through Internet and arrive at TCP/IP stack at remote computer.
 2. Characters are then delivered to OS and which later on passed to TELNET server.
 3. TELNET server changes that characters to characters which can be understandable by remote computer.
 4. Remote OS receives character from a pseudo-terminal driver, which is a piece of software that pretends that characters are coming from a terminal.
 5. Operating system on remote computer then passes character to the appropriate application program.

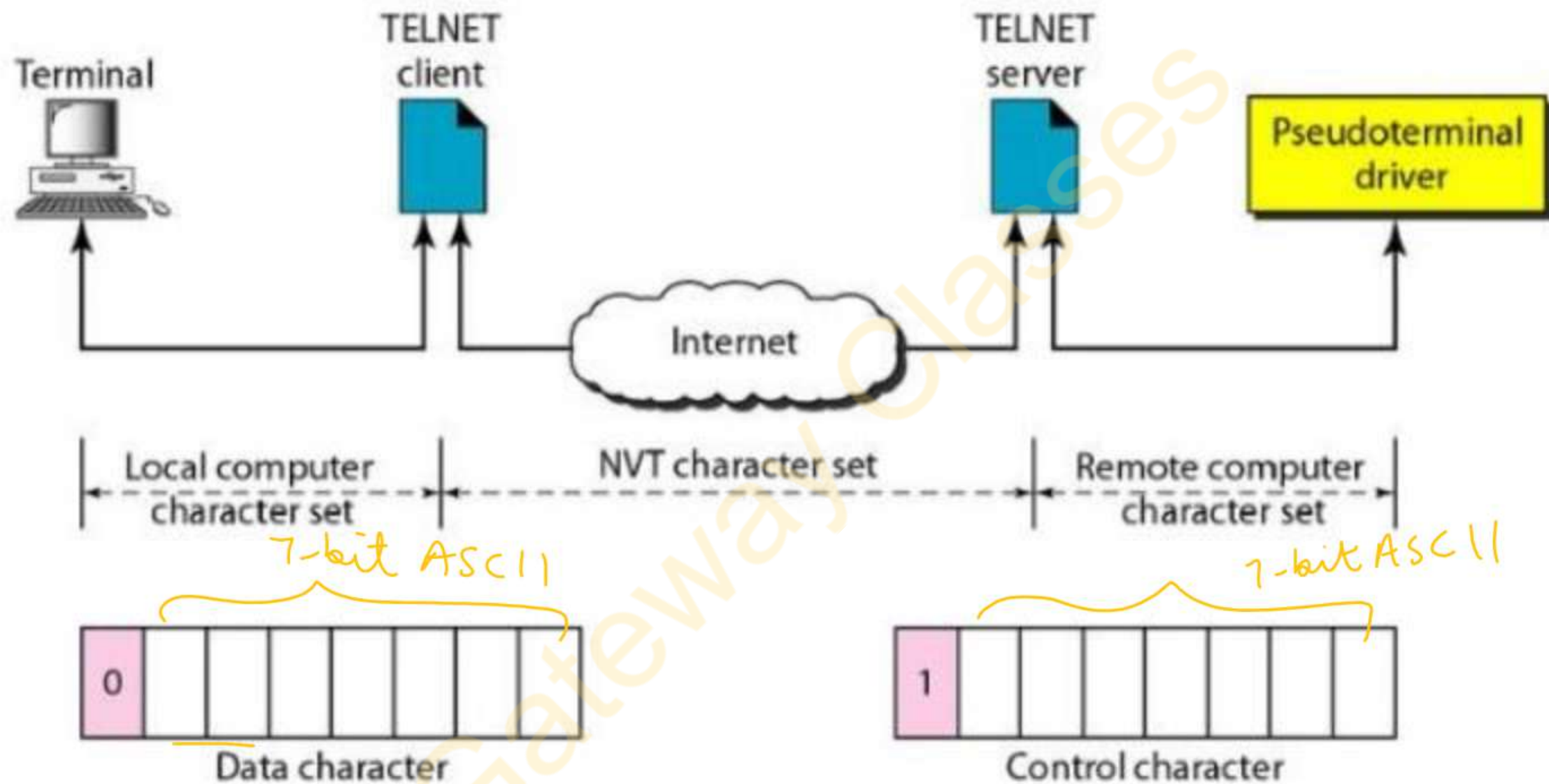
Procedure of Remote Login



NVT Character Set :

- With NVT Character set, TELNET client translates characters into NVT form and deliver to network.
- TELNET server translates data/commands from NVT form to form understandable by remote computer.
- NVT uses 2 sets of characters, one for data and other for control. Size of both characters is 8-bit bytes.
- **For data:**
 - NVT is an 8-bit character set in which 7 lowest bits are same as ASCII and highest order bit is 0.
- **For control characters:**
 - NVT uses an 8-bit character set in which the highest bit is set to 1.

Concept of NVT



TELNET Mode of Operation

Default Mode:

- Used if no other modes are invoked through option negotiation.
- The user types a character, and the client echoes the character on screen but does not send it until a whole line is completed.

Character Mode:

- Each character typed is sent by the client to the server.
- The server normally echoes the character back to be displayed on the client screen.
- Echoing of the character can be delayed if the transmission time is long.

Line Mode:

- Compensate for the deficiencies of the default mode and the character mode.
- Line editing (echoing, character erasing, line erasing) is done by client. The client then sends the whole line to the server.

Electronic Mail

It allows a message to include text, audio, and video. It also allows one message to be sent to one or more recipients.

First Scenario:

- Sender and receiver of e-mail are users (or application programs) on the same system.
- Admin created one mailbox on local hard drive for each user with permission restrictions where the received messages are stored.
- When Alice needs to send a message to Bob, Alice runs a user agent program to prepare the message and store it in Bob's mailbox.
- The message has the sender and recipient mailbox addresses. Bob can retrieve and read the contents of his mailbox using a user agent.

When the sender and the receiver of an e-mail are on the same system, we need only two user agents.

Second Scenario:

Sender and receiver of the e-mail are users (or application programs) on two different systems.

The message needs to be sent over the Internet.

Here we need user agents and message transfer agents.

Alice use user agent to send her message to system(mail server) at her own site.

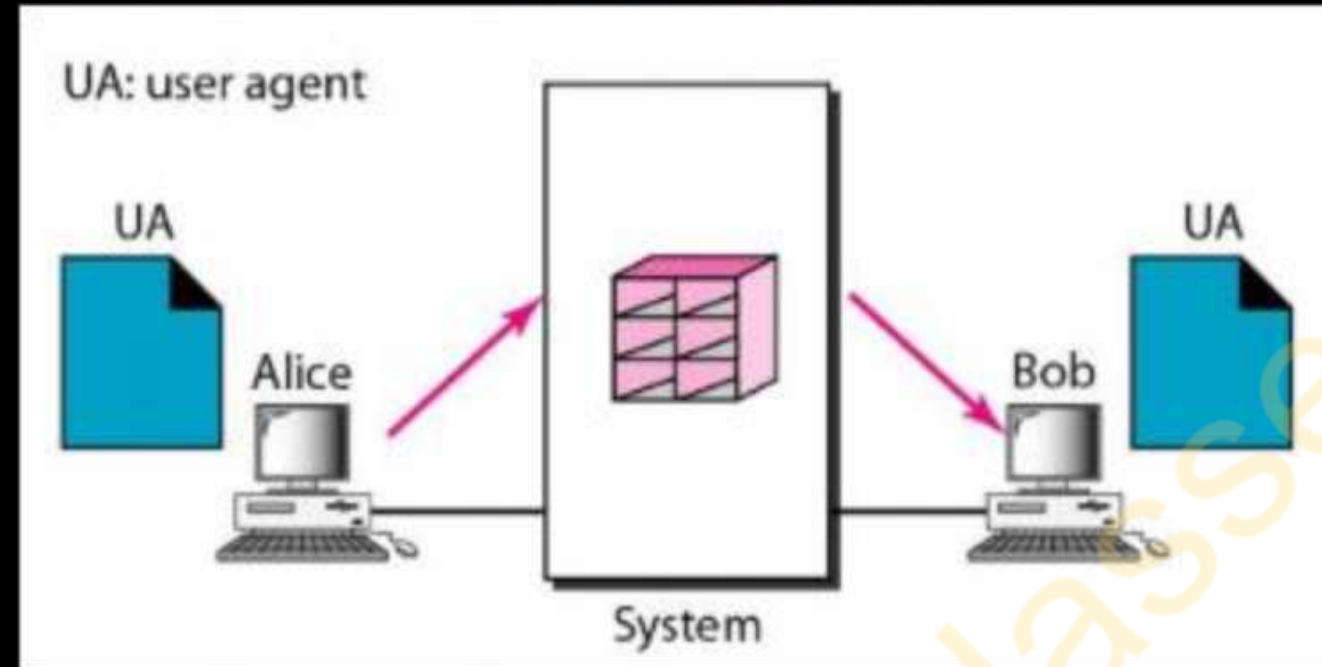
Mail server at her site uses a queue to store messages waiting to be sent.

Bob also needs a user agent to retrieve messages stored in the mailbox of the system at his site.

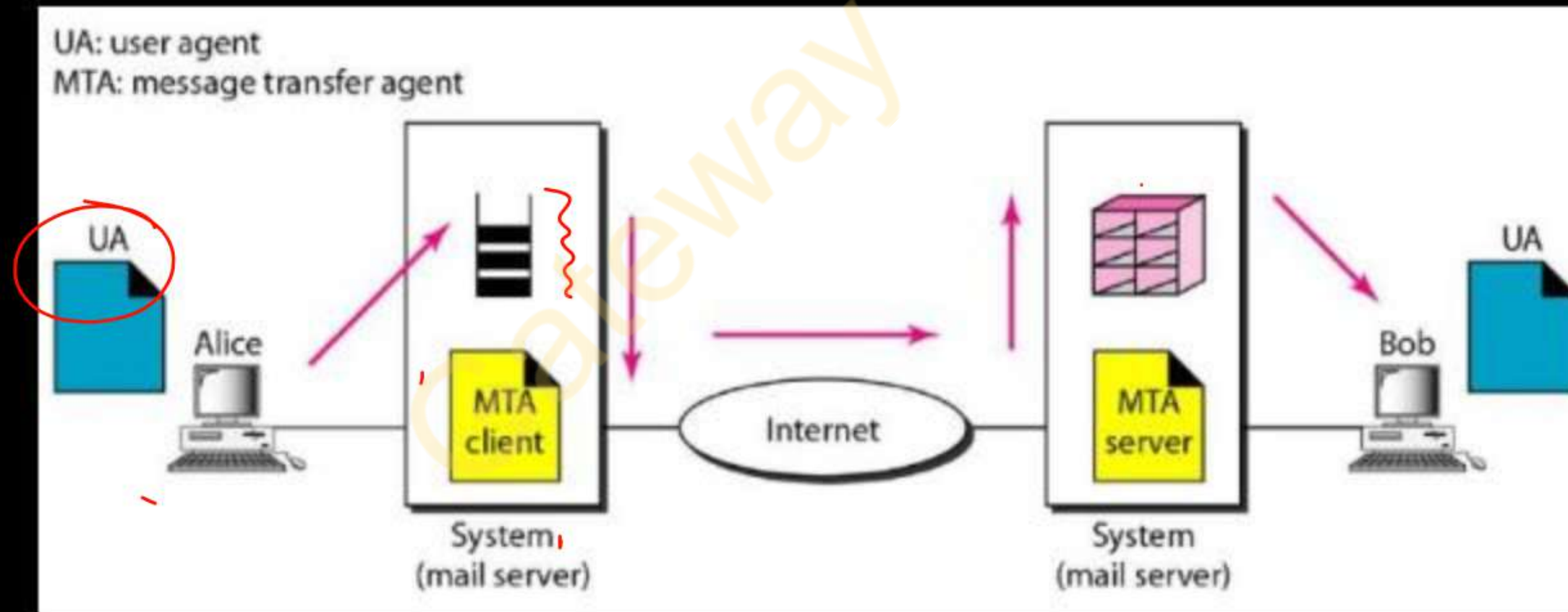
The message needs to be sent through the Internet from Alice's site to Bob's site.

When the sender and the receiver of an e-mail are on different systems, we need two UAs and a pair of MTAs (client and server).

First Scenario



Second Scenario

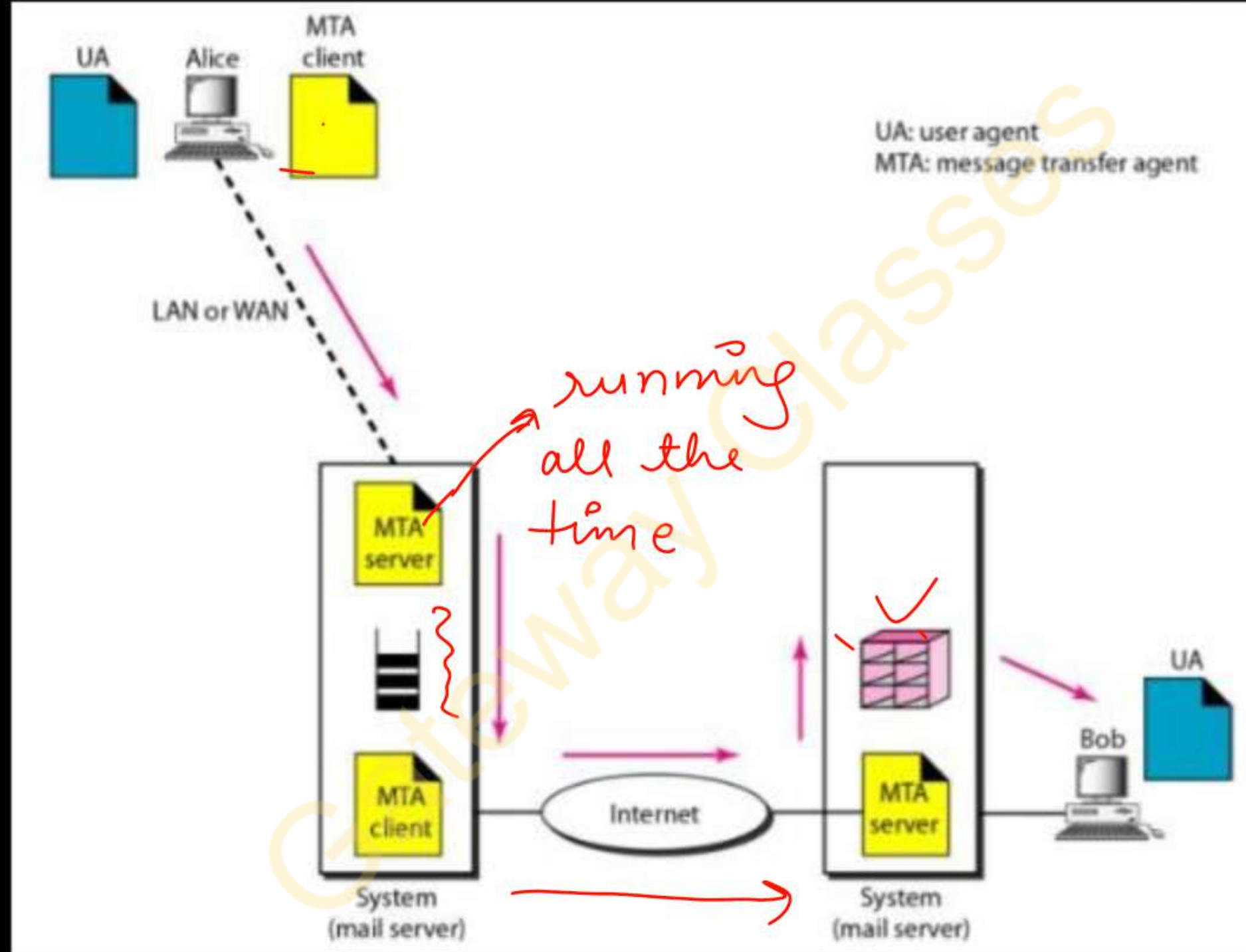


Third Scenario:

- Bob is directly connected to his system. Alice is separated from her system.
- Either Alice is connected to the system via a P2P WAN, or to a LAN in an organization that uses one mail server for handling e-mails.
- Whenever Alice has a message to send, she calls UA which calls MTA client.
- The MTA client connect with MTA server on the system, which is running all the time.
- The system at Alice's site queues all messages received. It then uses an MTA client to send the messages to the system at Bob's site; the system receives the message and stores it in Bob's mailbox.
- Bob uses his UA to retrieve message and reads it.

When the sender is connected to the mail server via a LAN or a WAN,
we need two UAs and two pairs of MTAs (client and server).

Third Scenario

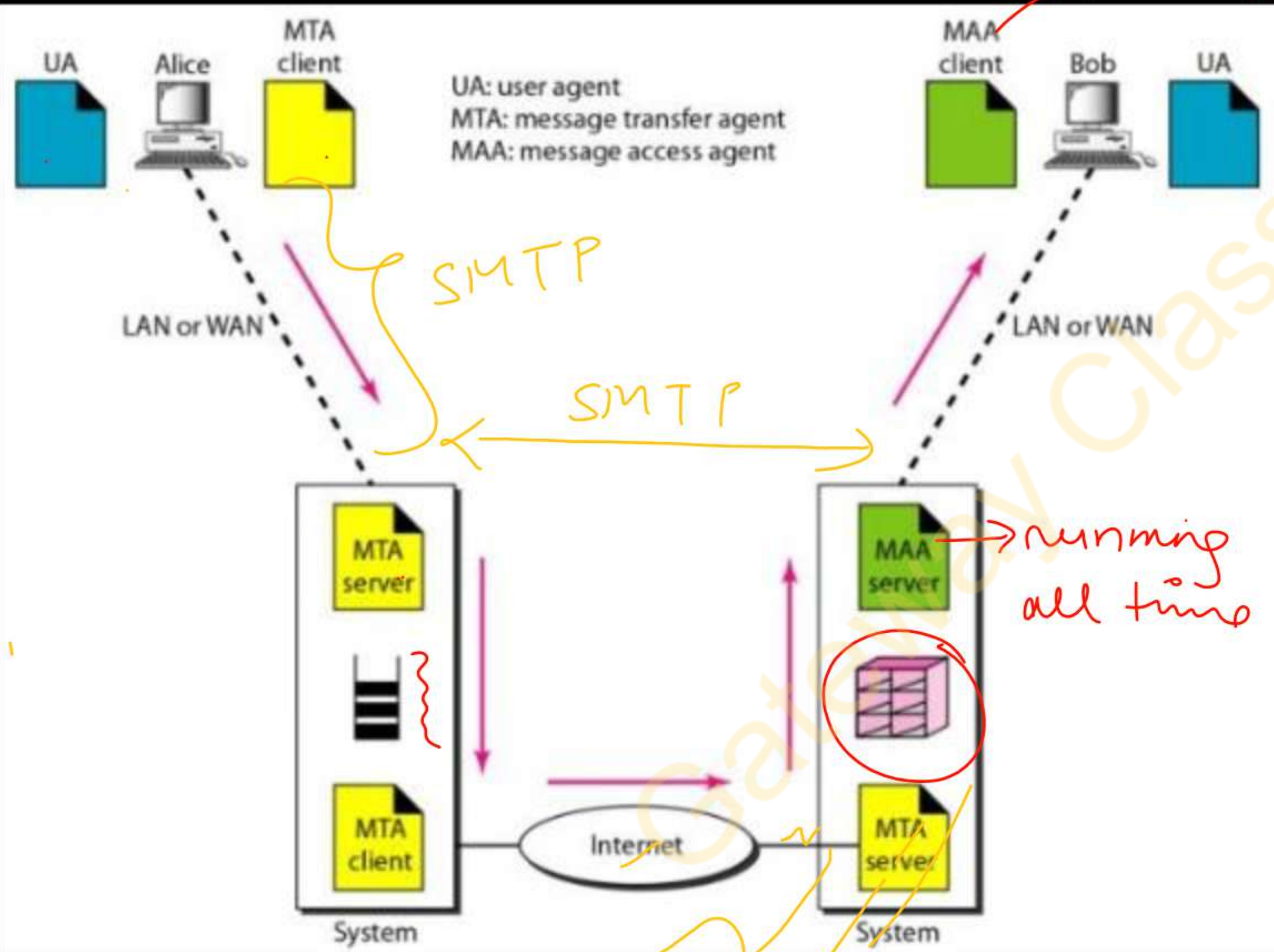


Fourth Scenario: This is the most common situation today.

- Bob is also connected to his mail server by a WAN or a LAN.
- Bob needs to retrieve message at Bob's mail server using message access Agent(MAA) client.
- The client sends a request to MAA server(running all time), and requests the transfer of messages.
- Bob cannot bypass the mail server and use the MTA server directly.
- Bob needs another pair of client/server programs: message access programs.
- This is so because an MTA client/server program is a push program:
 - the client pushes the message to the server. Bob needs a pull program.
 - The client needs to pull the message from the server.

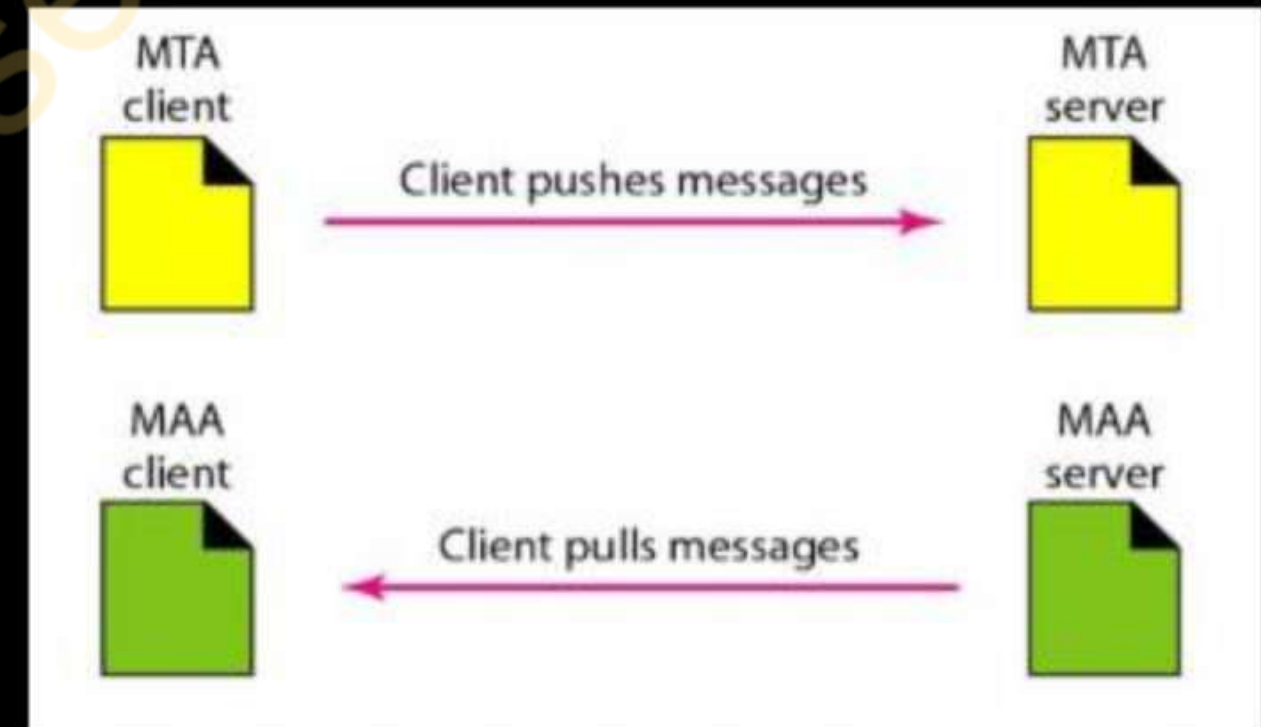
When sender and receiver are connected to mail server via LAN/ WAN, we need two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server).

Fourth Scenario



Message Access Agent

Push versus pull in electronic mail



User Agent (UA): First component of an electronic mail system.

Services Provided by UA: A software package that composes, reads, replies to, and forwards messages.

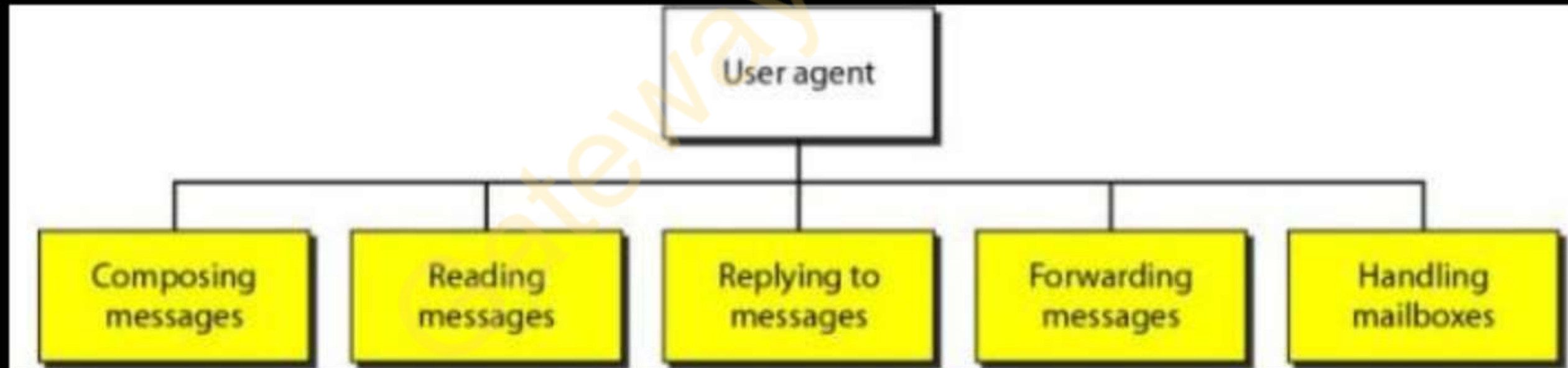
It also handles mailboxes.

1. **Composing Messages:** A user agent helps the user compose the e-mail message to be sent out.
 - Most user agents provide a template on the screen to be filled in by the user.
 - Some even have a built-in editor that can do spell checking, grammar checking or user can use his favourite text editor or word processor to create the message.
2. **Reading Messages:** Read incoming messages. When user invokes UA, it first checks mail in incoming mailbox. Shows a one-line summary of each received mail.

Each e-mail contains:

1. A number field.
2. A flag field that shows the status of the mail such as new, already read but not replied to, or read and replied to.
3. size of message.
4. sender.
5. Optional subject field.

3. **Replying to Messages:** After reading a message, a user can use UA to reply to a message.
- UA usually allows user to reply to original sender or to reply to all recipients of the message.
 - The reply message may contain the original message (for quick reference) and the new message.
4. **Forwarding Messages:**
- Sending the message to a third party.
 - UA allows the receiver to forward the message, with or without extra comments, to a third party.



User Agent Types

1. Command-driven UA:

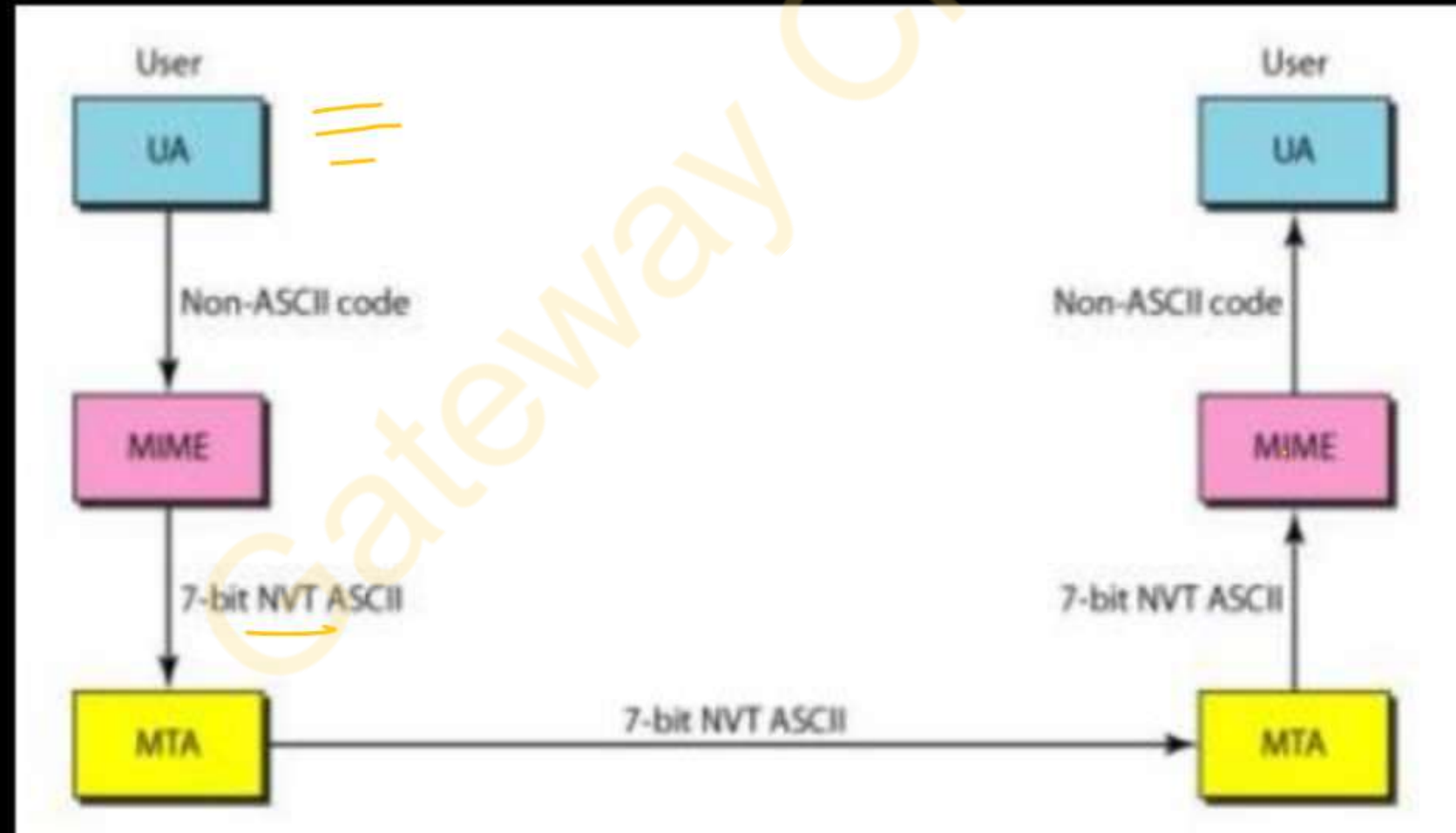
- Belong to early days of Email.
- Normally accepts a one-character command from keyboard to perform its task.
- A user can type the character r, at the command prompt, to reply to the sender of the message, or type the character R to reply to the sender and all recipients. **Example:** mail, pine, and elm.

2. GUI-Based Modem UA:

- Contain graphical-user interface components that allow the user to interact with the software by using both the keyboard and the mouse.
- They have graphical components such as icons, menu bars, and windows that make the services easy to access. **Example:** Eudora, Outlook, and Netscape.

Multipurpose Internet Mail Extensions (MIME)

- Email can send messages only in NVT 7-bit ASCII format.
- So, cannot be used for languages such as French, German etc. and to send binary/video/audio data.
- Multipurpose Internet Mail Extensions (MIME) allows non-ASCII data to be sent through e-mail.
- MIME transforms non-ASCII data at sender site to NVT ASCII data and delivers them to client MTA to be sent through Internet. The message at the receiving side is transformed back to the original data.



Message Transfer Agent: SMTP

(Imp)

- The actual mail transfer is done through message transfer agents. (MTAs)
- To send mail, a system must have the client MTA, and to receive mail, a server MTA.
- The protocol that defines the MTA client and server in the Internet is called the Simple Mail Transfer Protocol (SMTP).

SMTP is used two times:

- between the sender and the sender's mail server
- between the two mail servers.

SMTP simply defines how commands and responses must be sent back and forth.

Each network is free to choose a software package for implementation.

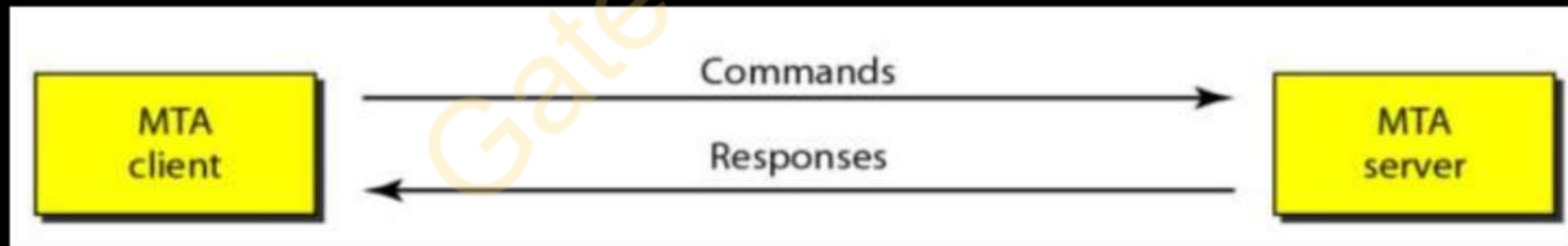
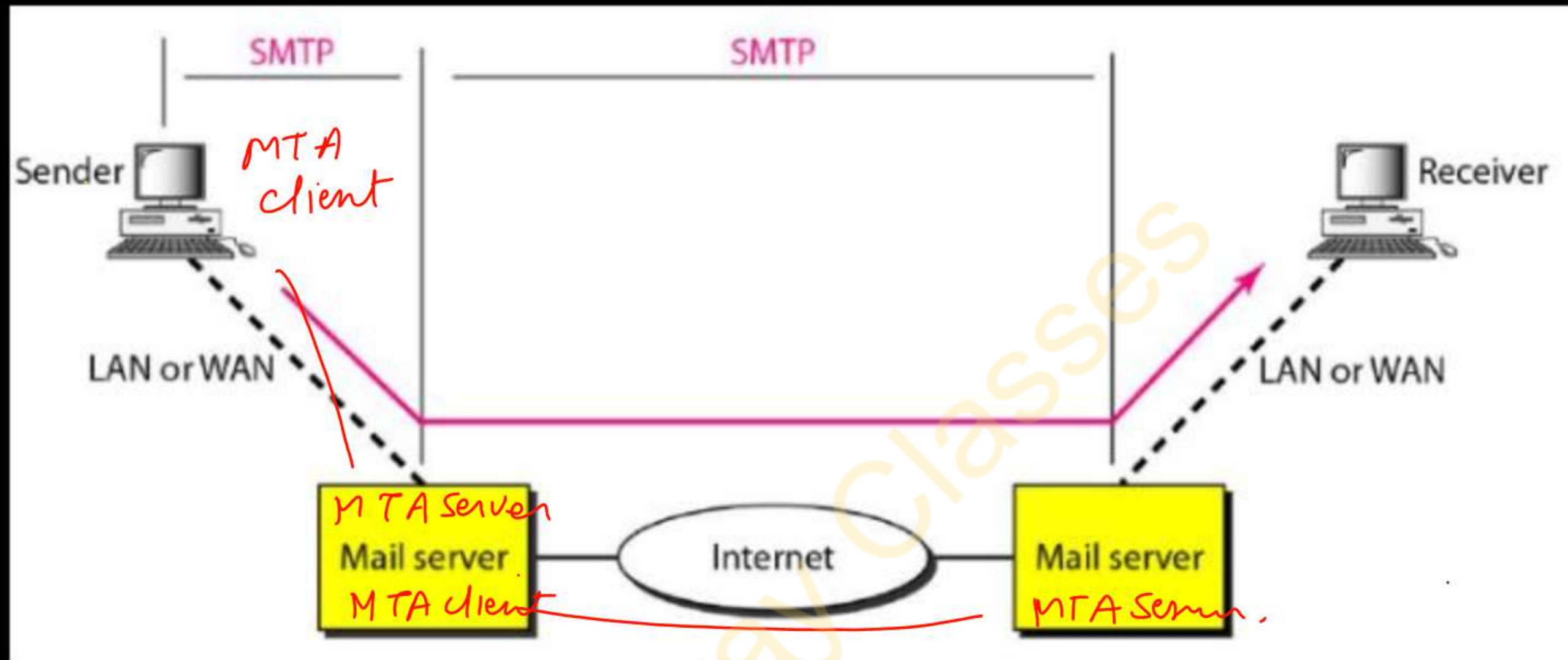
Simple Mail Transfer Protocol (SMTP)

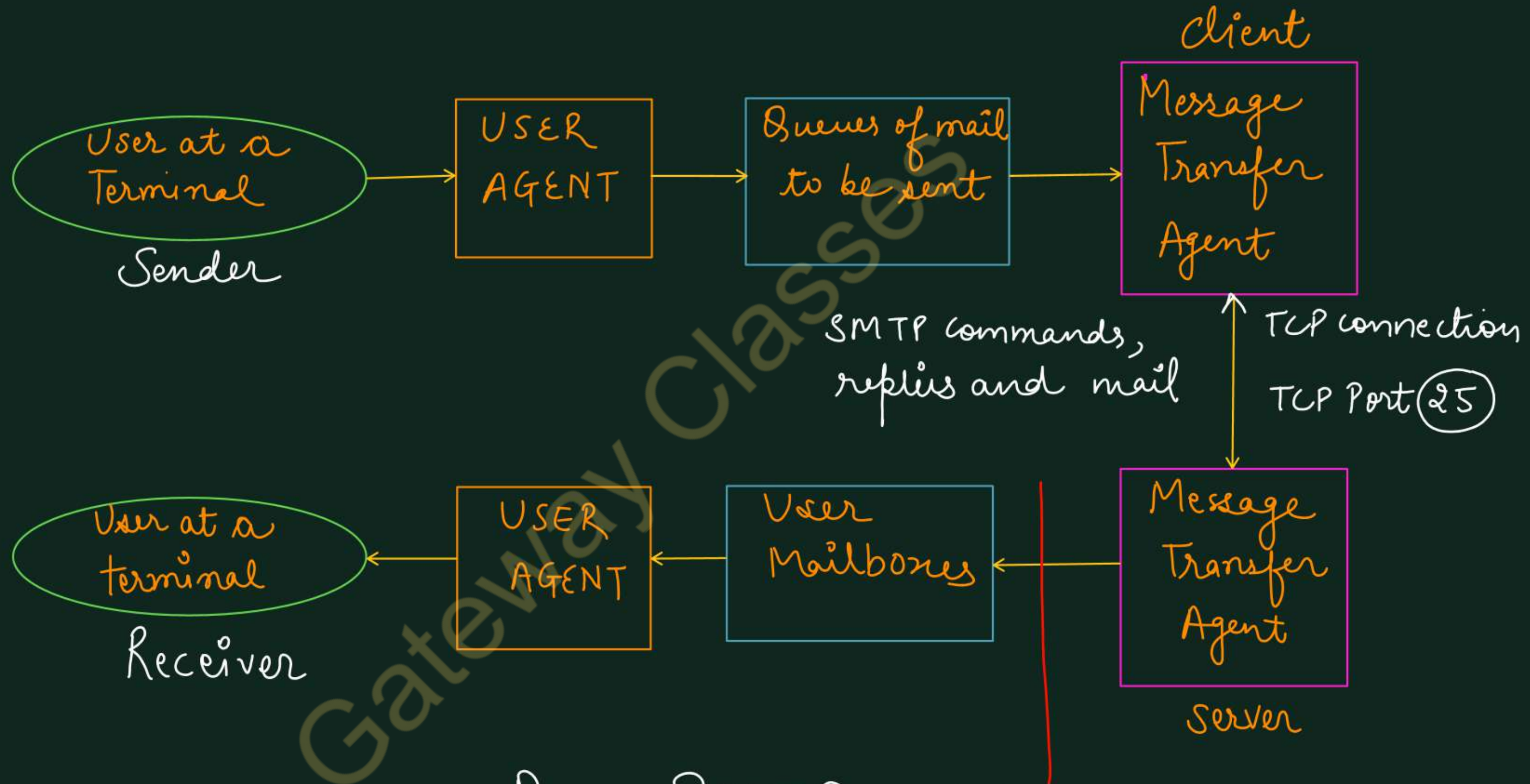
- Application layer protocol used for exchanging email messages between servers.
- To send an email, the client opens a TCP connection to the SMTP server. The server, which is always listening on port 25, initiates the connection as soon as it detects a client. Once the TCP connection is established, the client sends the email across the connection.

Types of SMTP Protocol

Supports two types of email delivery methods: end-to-end and store-and-forward.

1. **End-to-end**: Email is sent directly from the sender's SMTP client to the recipient's SMTP server without passing through intermediate servers.
2. **Store-and-forward**: Email may pass through several intermediate servers (Message Transfer Agents, or MTAs) before reaching the recipient.





BASIC SMTP model

How does SMTP Work?

1. Sending Email:

- When a user wants to send an email, they use a User Agent (UA), like Outlook or Gmail.
- The email is handed over to the MTA, which transfers the email to the recipient's mail server.

2. SMTP Client and Server:

- Sender-SMTP (Client): Email sender's MTA initiates connection to recipient's MTA (Receiver-SMTP).
- Receiver-SMTP (Server): Receiving MTA listens for incoming connections and receives email from sender-SMTP.
- This communication happens over TCP port 25.

3. Relays and Gateways:

- Relays: In some cases, the email may pass through several intermediate MTAs before reaching the destination server. These MTAs act as relays.
- Gateways: If the sending and receiving systems use different email protocols (e.g., SMTP and non-SMTP), an email gateway can convert the email to the appropriate format for delivery.

4. Email Delivery:

- The sender's MTA sends the email to the receiver's MTA, either directly or through relays.
- MTA uses SMTP protocol to transfer the message. Once it's delivered to the destination MTA, the email is placed in the recipient's mailbox.
- The recipient's User Agent can then download the email.

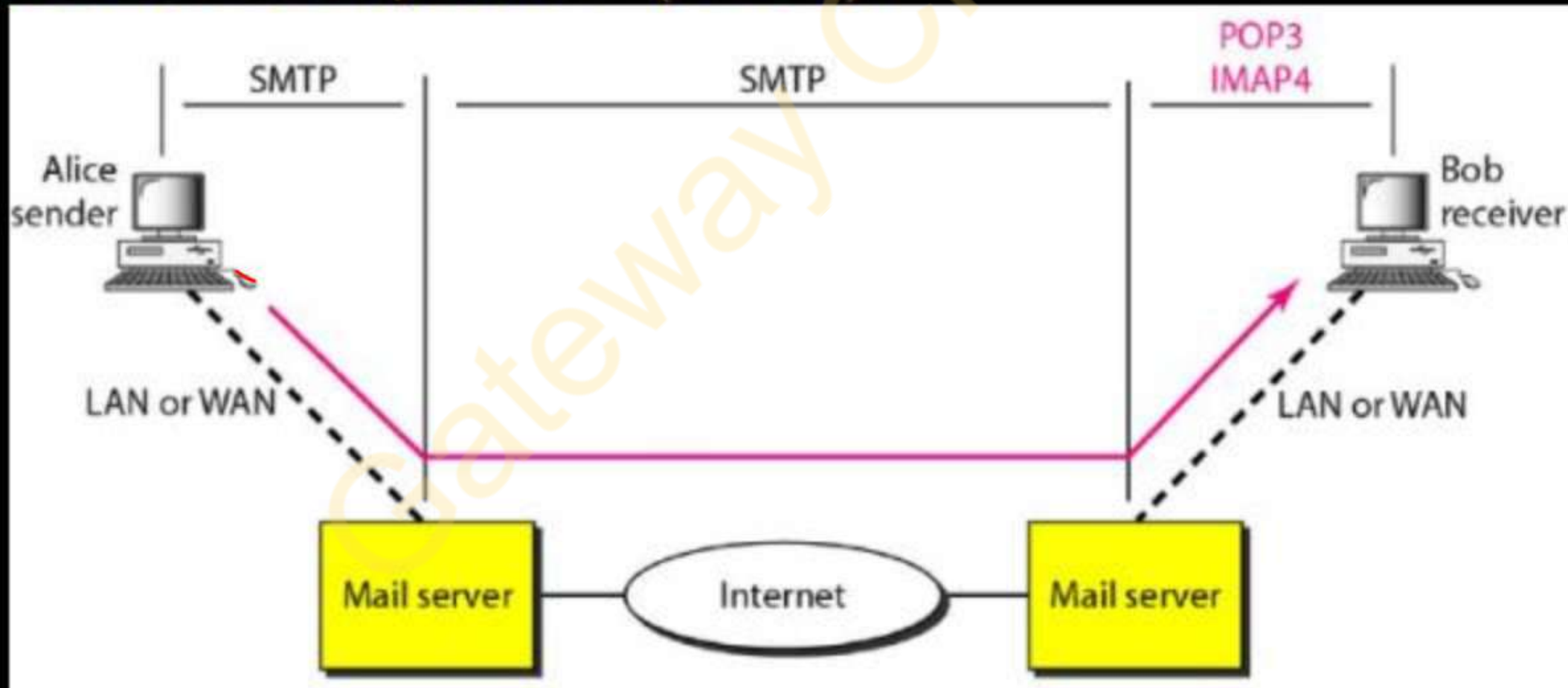
Message Access Agent: **POP** and IMAP

SMTP is a *push* protocol; it pushes the message from the client to the server.

Now, the client must pull messages from the server so the third stage uses a message access agent.

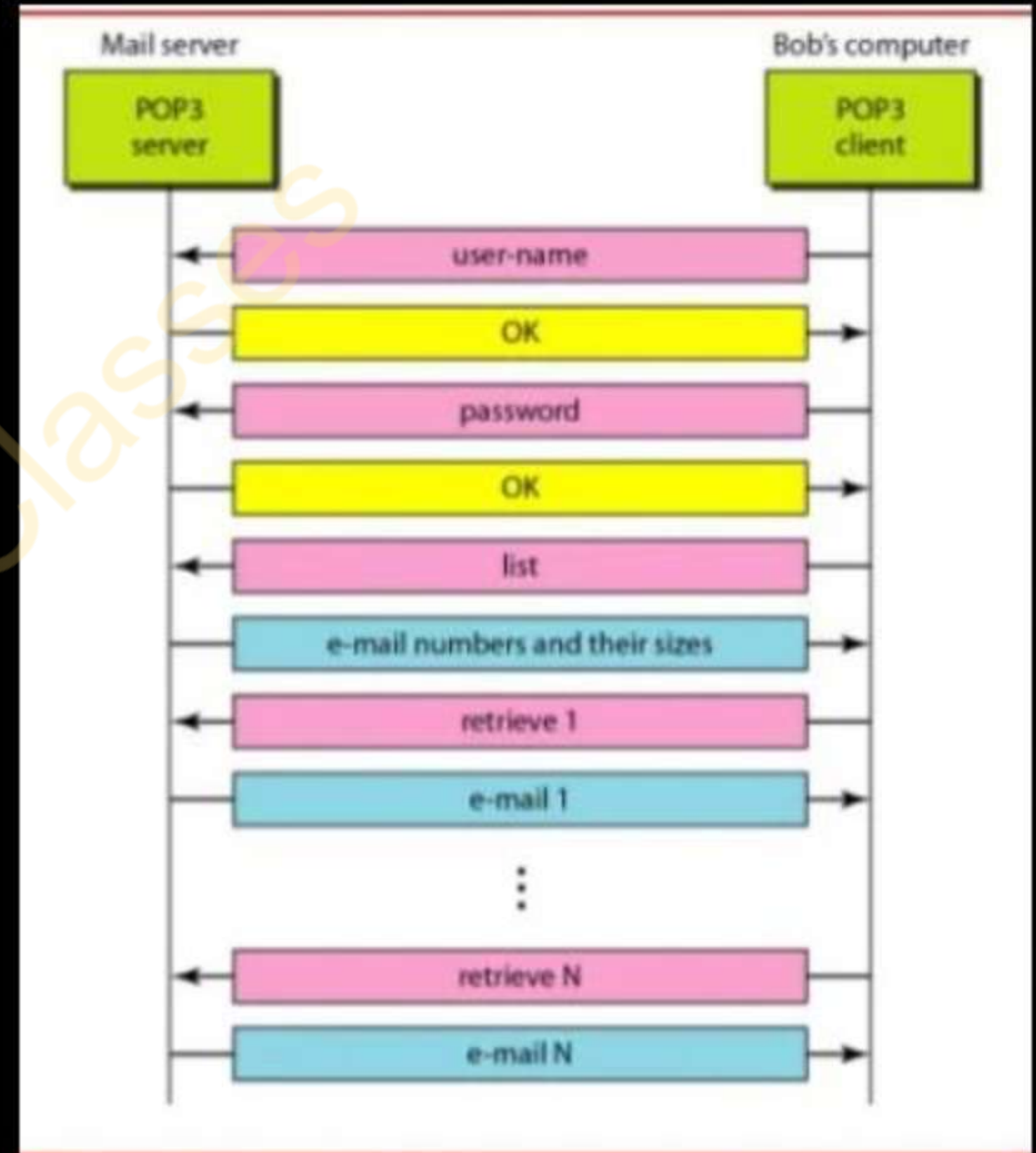
Currently two message access protocols are available:

1. **Post Office Protocol, version 3 (POP3)**
2. **Internet Mail Access Protocol, version 4 (IMAP4)**



POP3:

- Client POP3 software is installed on recipient computer; server POP3 software on the mail server.
- when user needs to download e-mail from mailbox on mail server, client opens a connection to the server on TCP port 110. It then sends its user name and password to access the mailbox. The user can then list and retrieve the mail messages, one by one.



POP3 has two modes: the delete mode and the keep mode.

- In delete mode, the mail is deleted from the mailbox after each retrieval.
- In keep mode, the mail remains in the mailbox after retrieval.

Problems:

- Does not allow user to organize his mail on server; user cannot have different folders on server.
- Does not allow the user to partially check the contents of the mail before downloading.

IMAP4 provides the following extra functions:

- A user can check the e-mail header prior to downloading.
- A user can search the contents of the e-mail for a specific string of characters prior to downloading.
- A user can partially download e-mail. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
- A user can create, delete, or rename mailboxes on the mail server.
- A user can create a hierarchy of mailboxes in a folder for e-mail storage.

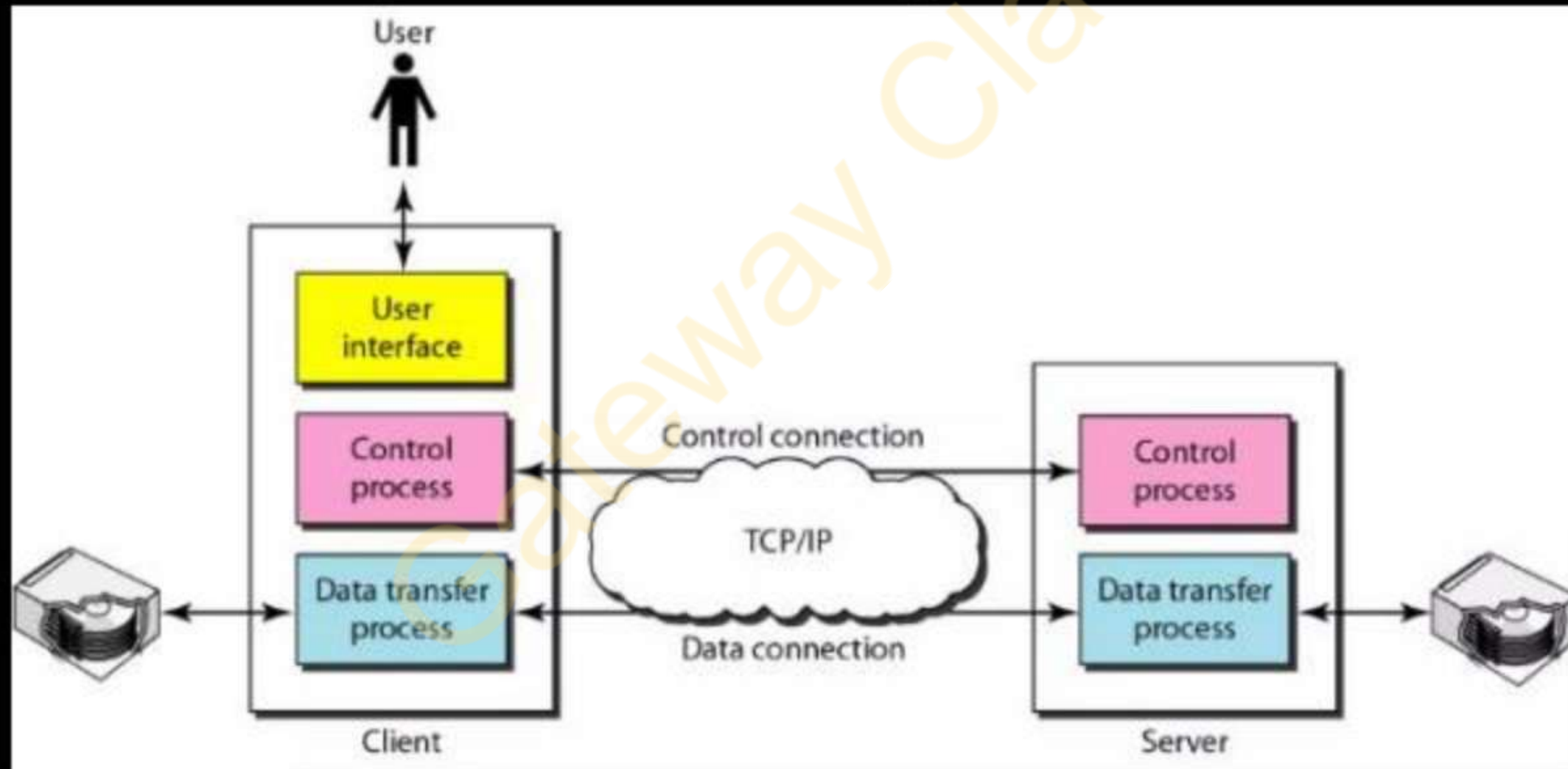
File Transfer Protocol (FTP) (Imp)

- Standard mechanism provided by *TCP/IP* for copying a file from one host to another.
- If two systems may use different file name conventions, different ways to represent text and data, different directory structures etc.
- All these problems have been solved by FTP.

It uses TCP services and needs two TCP connections:

- One connection is used for data transfer,
 - Other for control information (commands and responses).
-
- In control connection we transfer only a line of command or line of response at a time. (less complex)
 - In data connection, variety of data types are transferred. (More complex) (handled at FTP level).
- FTP uses two well-known TCP ports at server side: Port 21 for control connection, and port 20 for data connection.

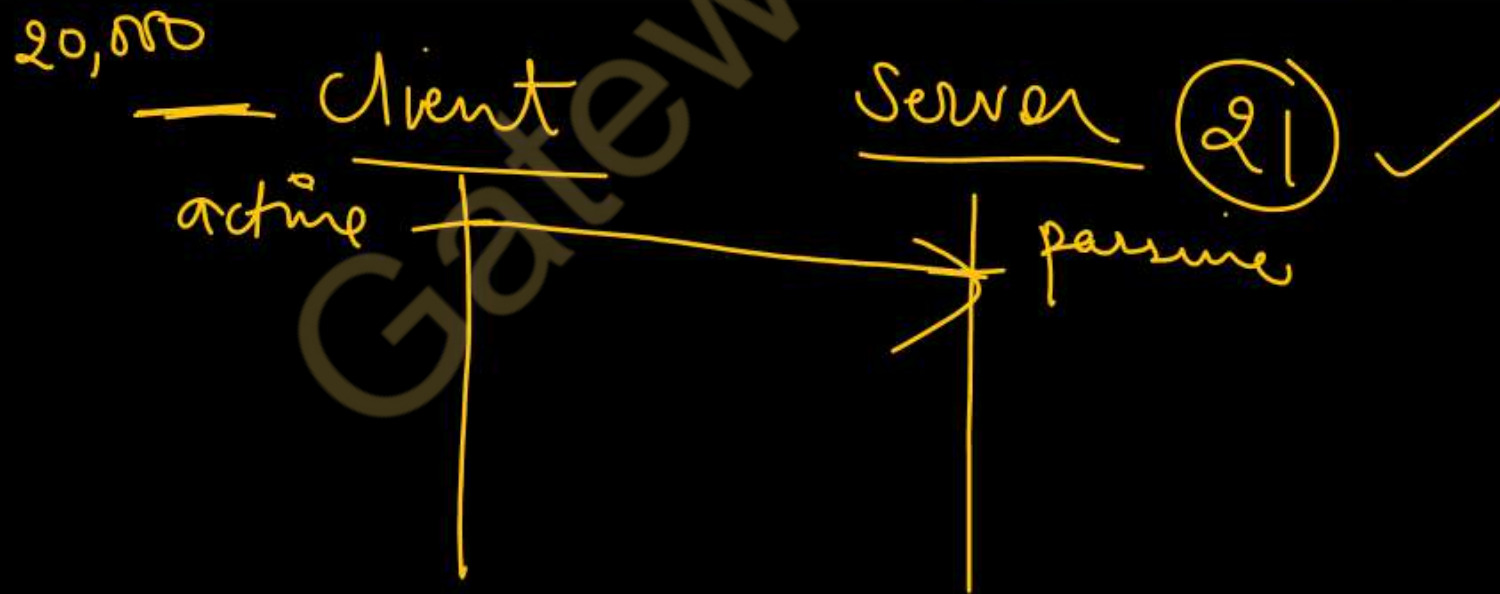
- **Control connection** is made between control processes and remains connected during entire interactive FTP session. **Data connection** is made between data transfer processes. It is opened and then closed for each file transferred.
- When user starts an FTP session, the control connection opens. While the control connection is open, the data connection can be opened and closed multiple times if several files are transferred.



Steps to establish a control connection:

1. The server issues a passive open on the well-known port 21 and waits for the client.
2. After server issues passive open, the client issue active open using an ephemeral port.
3. Client sends login username and password over the control connection to gain access.
4. Once authenticated, the client sends commands (e.g., ls, cd, get, put) over the control connection to navigate directories, list files, and request file transfers.

This control connection remains open throughout the process.



Steps to establish a Data connection:

1. **Mode selection:** FTP operating in active or passive mode, affect the way data connection is set up.

Active mode:

- Client initiates a data connection to the server on a specific port.
- Client sends the PORT command to inform the server of the port it is listening on.
- Server issues active open using well-known port 20 and initiates a data connection to client on the port informed by the client.

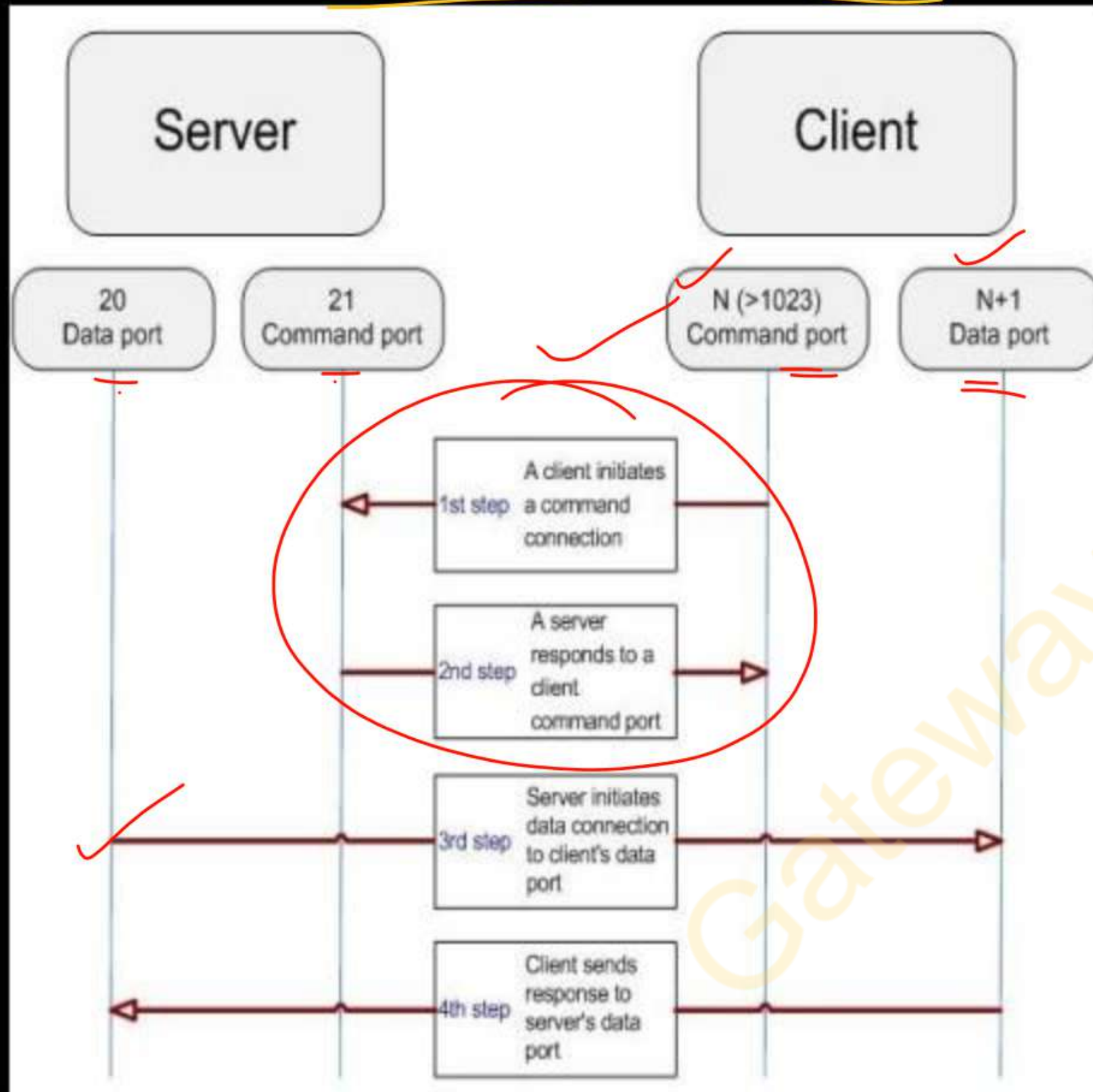
Passive mode:

- The client sends the PASV command to the server.
- The server listens on a specific port for the client's data connection.
- The client then initiates a data connection to the server.

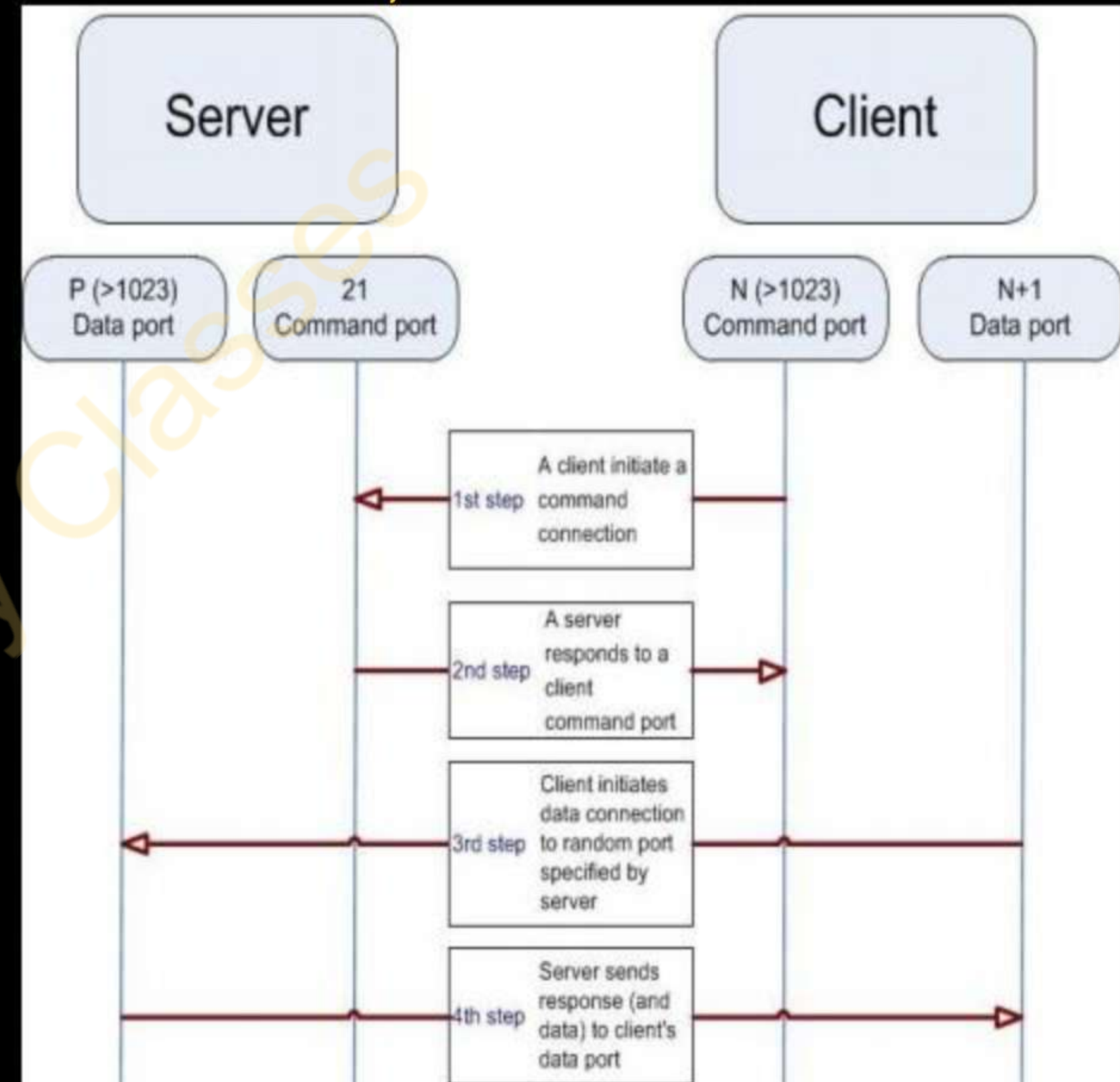
2. **Data transfer:** file transfer (upload or download) occurs over this connection.

3. **Connection closure:** The data connection is closed after each file transfer.

Active FTP Connection



Passive FTP connection



Type of FTP Connections: FTP connections are of two types:

1. Active FTP connection:

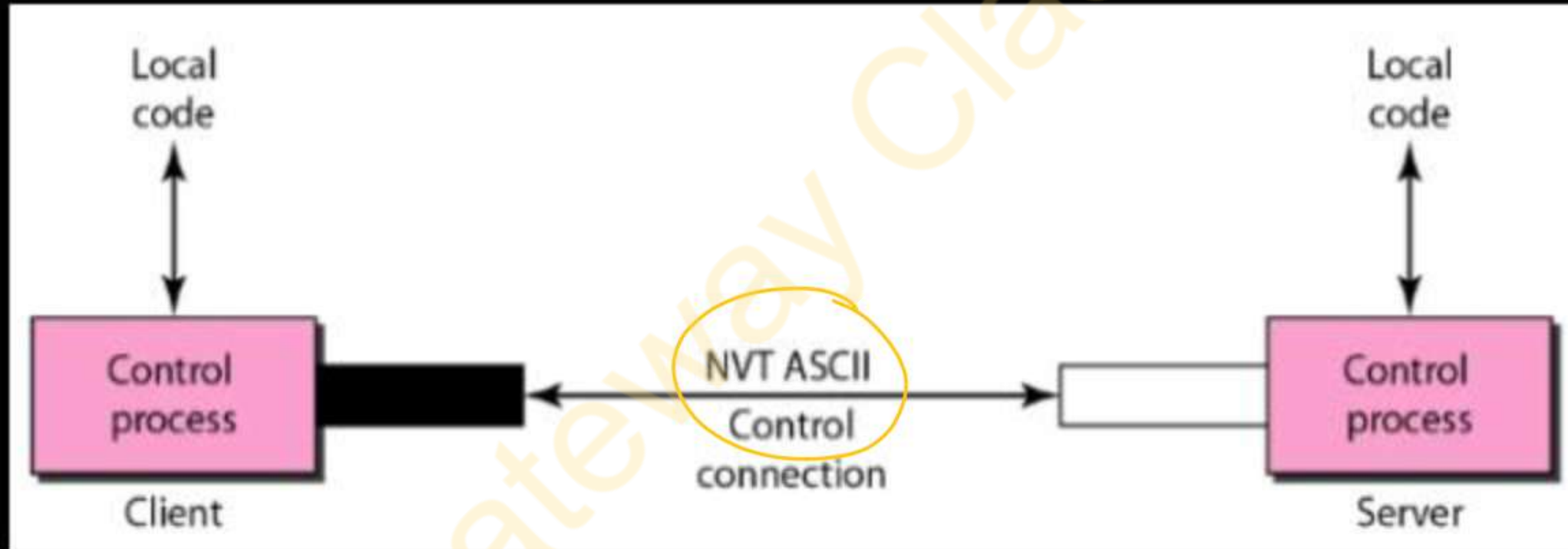
- Client establishes command channel and server establishes data channel.
- When client requests data over the connection, server initiates the transfer of data to the client.
- It is not a default connection and cause problems if there is firewall between client and server.

2. Passive FTP connection(default connection):

- Client establishes both data channel as well command channel.
- When client requests data over the connection, server sends a random port number to the client.
- As soon as client receives this port number it establishes the data channel.
- It is the default connection, as it works better even if the client is protected by the firewall.

Communication over Control Connection:

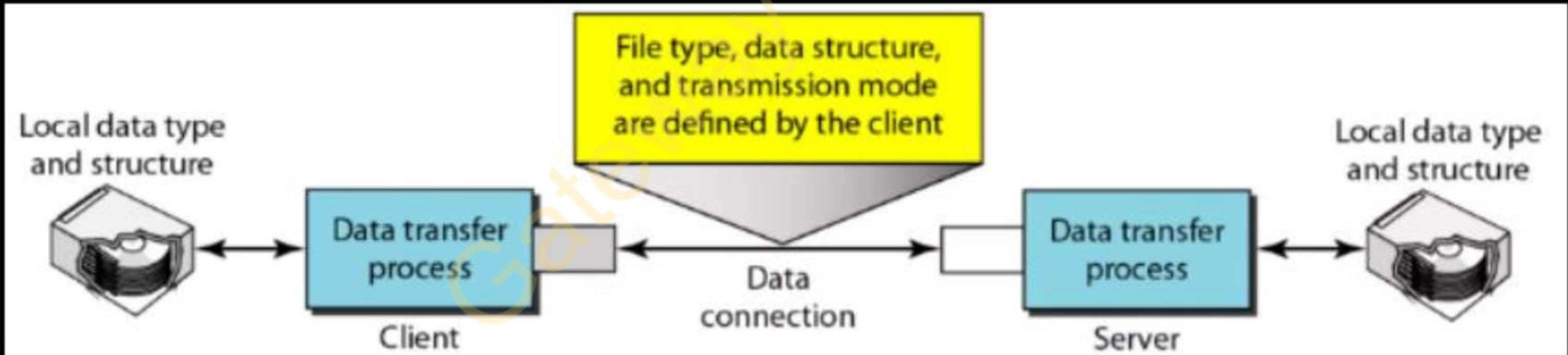
- It uses the 7-bit ASCII character set. Communication is achieved through commands and responses.
Each command or response is only one short line, so we need not worry about file format or structure.
- Each line is terminated with a two-character (carriage return and line feed) end-of-line token.



Communication over Data Connection:

File transfer occurs over data connection under the control of the commands sent over the control connection. File transfer in FTP means one of three things:

1. A file is to be copied from server to client. This is called retrieving a file (RETR command).
2. A file is to be copied from client to server. This is called storing a file (STOR command).
3. A list of directory or file names is to be sent from server to client (LIST command).



Why FTP?

- There are various other protocols like HTTP which are used to transfer files between computers, but they lack clarity and focus as compared to FTP.
- Systems involved in connection are heterogeneous systems, they differ in OS, directories, structures, character sets, etc. FTP shields user from these differences and transfers data efficiently and reliably.
- ✓ FTP can transfer ASCII, EBCDIC, or image files.
- The ASCII is the default file share format, in this, each character is encoded by NVT ASCII. - 7-bit
- In ASCII or EBCDIC, the destination must be ready to accept files in this mode.
- The image file format is the default format for transforming binary files.

File Transmission Mode:

FTP transfer files using any of the following modes:

1. **Stream Mode(default mode):** data is transferred from FTP to TCP in stream of bytes. TCP fragments data into small segments. The connection is automatically closed if the transforming data is already in form of stream of bytes. Otherwise, the sender will close the connection.
2. **Block Mode:** Data is transferred from FTP to TCP in the form of blocks, and each block followed by a 3-byte header. The first byte of block(description block) contains the info about block and the other two bytes contain the size of the block.
3. **Compressed Mode:** Used to transfer big files. Used to decrease the size of the file into small and send it on the internet.

Anonymous FTP

- Some sites can enable anonymous FTP whose files are available for public access. So, the user can access those files without any username or password.
- Access of user is very limited such as the user can copy the files but not allowed to navigate through directories.

Disadvantages of FTP:

- Lack of Built-in Security
- Firewall Issues
- Complex Configuration
- Not Ideal for Real-Time Data Transfer

TFTP (Trivial File Transfer Protocol):

- A network protocol designed for **simple file transfers between devices within a local network.**
- Simpler FTP designed for transferring small files which do not require the added complexity such as log on credentials or encryption.
- TFTP runs in a minimalist mode and **well adapted to applications that require only simple file transfers without much interaction.**
- It relies on using the UDP and communicates over port 69.
- Simplicity limits the security and error-checking features associated with TFTP, and it is used in secure, **closed network environments rather than on the internet.**
- It is faster, but less stable.
- Frequently used when little setup is needed.

Difference Between FTP and TFTP (Imp) AKTU

Feature	FTP	TFTP
Connection	Establishes connection between two computers, allowing more complex set of commands and options	Establishes a connection between two computers with limited set of commands and options
Authentication	username and password for authentication	Does not support authentication
Security	Encrypts data transfer	Does not encrypt data transfer
Error handling	Can recover from errors during transfer	Does not have error recovery
File transfer mode	Supports ASCII and binary transfer modes	Only supports binary transfer mode
Transfer options	Resume interrupted transfers and setting transfer mode, type, and other options	Does not support any transfer options

Difference between FTP and HTTP (AKTU)

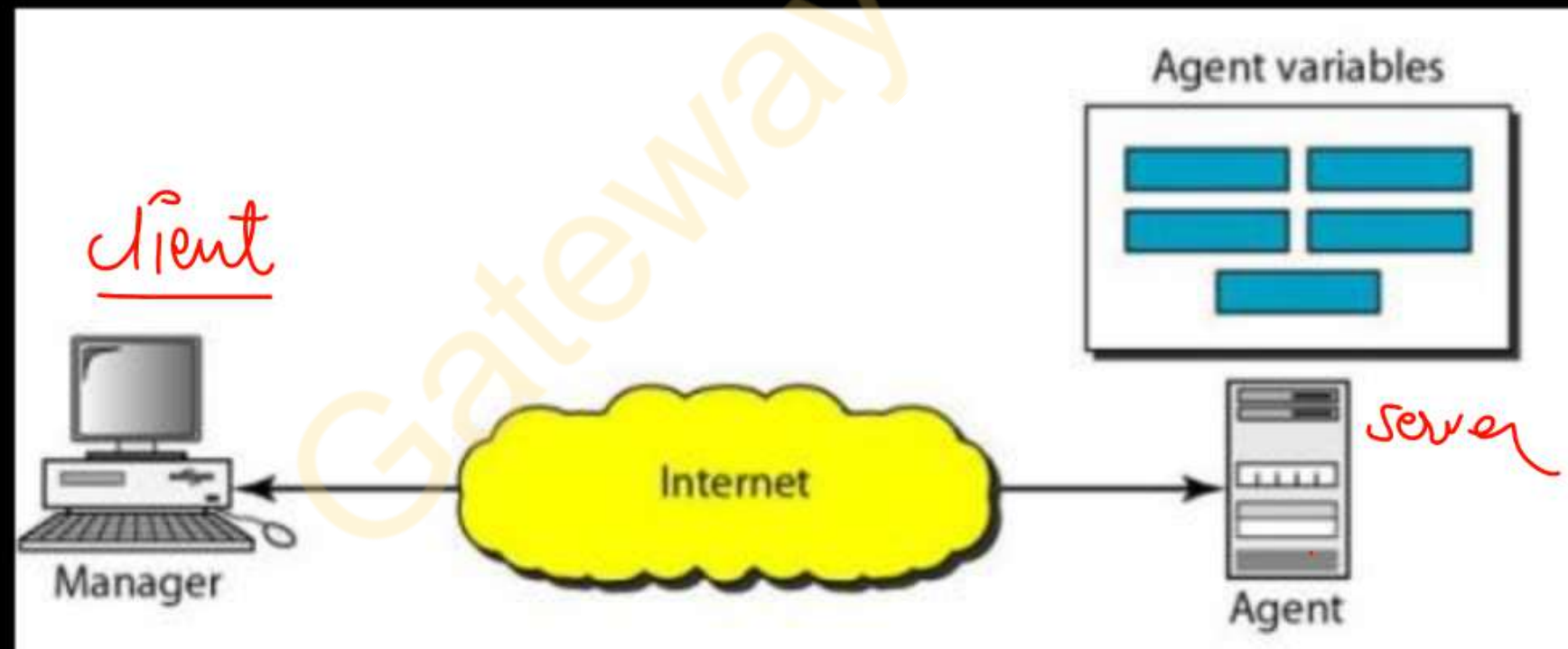
HTTP (HyperText Transfer Protocol)	FTP (File Transfer Protocol)
Set of rules to transfer web pages over internet.	Set of rules to download/upload files over internet.
Only supports data connection.	Supports both data connection and control connection
Uses TCP on port 80.	Uses TCP on port 20 and port 21.
Does not require authentication.	Requires authentication.
Efficient in transferring small files.	Efficient in transferring large files.
Files transferred over internet are not saved to memory.	Files transferred over internet are saved to the memory.
It is a stateless protocol.	It is not a stateless protocol and it maintains states.
Can use Persistent and Non-persistent TCP connection.	Uses a Persistent TCP connection for Control connection and Non-persistent for Data Connection.
It uses one way communication system.	It uses two-way communication system.
HTTP is faster.	FTP is slower as compared to HTTP.

Simple Network Management Protocol (SNMP)

- SNMP is a framework for managing devices in an internet using the TCP/IP protocol suite.
- It provides a set of fundamental operations for monitoring devices made by different manufacturers and installed on different physical networks, and maintaining an internet.

Concept:

SNMP is an application-level protocol in which a few manager stations (usually hosts) control a set of agents (usually routers).



Managers and Agents:

- Manager is a host that runs the SNMP client program. Managed station(agent), is a router or host runs the SNMP server program.
- Management is achieved through simple interaction between a manager and an agent.
- Agent keeps performance information in a database. Manager has access to values in the database.

Example:

- Router can store in appropriate variables the number of packets received and forwarded.
- Manager can fetch and compare values of these variables to see if the router is congested or not.
- The manager can also make the router perform certain actions.

Example:

- i. Router periodically checks value of a reboot counter to see when it should reboot itself.
Manager can use this feature to reboot the agent remotely at any time.

- ii. The server program running on the agent can check the environment, and if it notices something unusual, can send a warning message, called a trap, to the manager.

Management with SNMP is based on three basic ideas:

1. A manager checks an agent by requesting information that reflects the behavior of the agent.
2. A manager forces an agent to perform a task by resetting values in the agent database.
3. An agent contributes to the management process by warning the manager of an unusual situation.

Management Components:

SNMP uses two protocols:

1. Structure of Management Information (SMI)
2. Management Information Base (MIB)

Management on Internet is done through cooperation of three protocols SNMP, SMI, and MIB

Role of SNMP:

- It defines format of the packet to be sent from a manager to an agent and vice versa.
- The packets exchanged contain the object (variable) names and their status (values). SNMP is responsible for reading and changing these values.

Role of SMI:

- To use SNMP, we need rules for naming objects (as objects in SNMP form a hierarchical structure), rules to define the type (simple or structured), size, range, and encoding of objects.
- We need these universal rules because we do not know the architecture of the computers that send, receive, or store these values. SMI defines these rules.
- **SMI is a collection of general rules to name objects and to list their types.** The association of an object with the type is not done by SMI.

Role of MIB:

- For each entity to be managed, **MIB define the number of objects.**
- **Name the objects** according to the rules defined by SMI.
- **Associate a type to each named object.**
- MIB creates a set of objects defined for each entity similar to database (mostly metadata in database, names and types without values).

MIB creates a collection of named objects, their types, and their relationships to each other in an entity to be managed.

An Analogy

1. Before we write a program, syntax of PL (such as C or Java) must be predefined.

➤ PL defines structure of variables (simple, structured, pointer) and naming rules.

➤ PL defines type of data to be used (integer, float, char, etc.).

int xi = 10;

In programming rules are defined by PL while in network management rules are defined by SMI

2. Most computer languages require variables be declared in each specific program.

20.

➤ As the types are predefined in PL, the program knows the range and size of each variable.

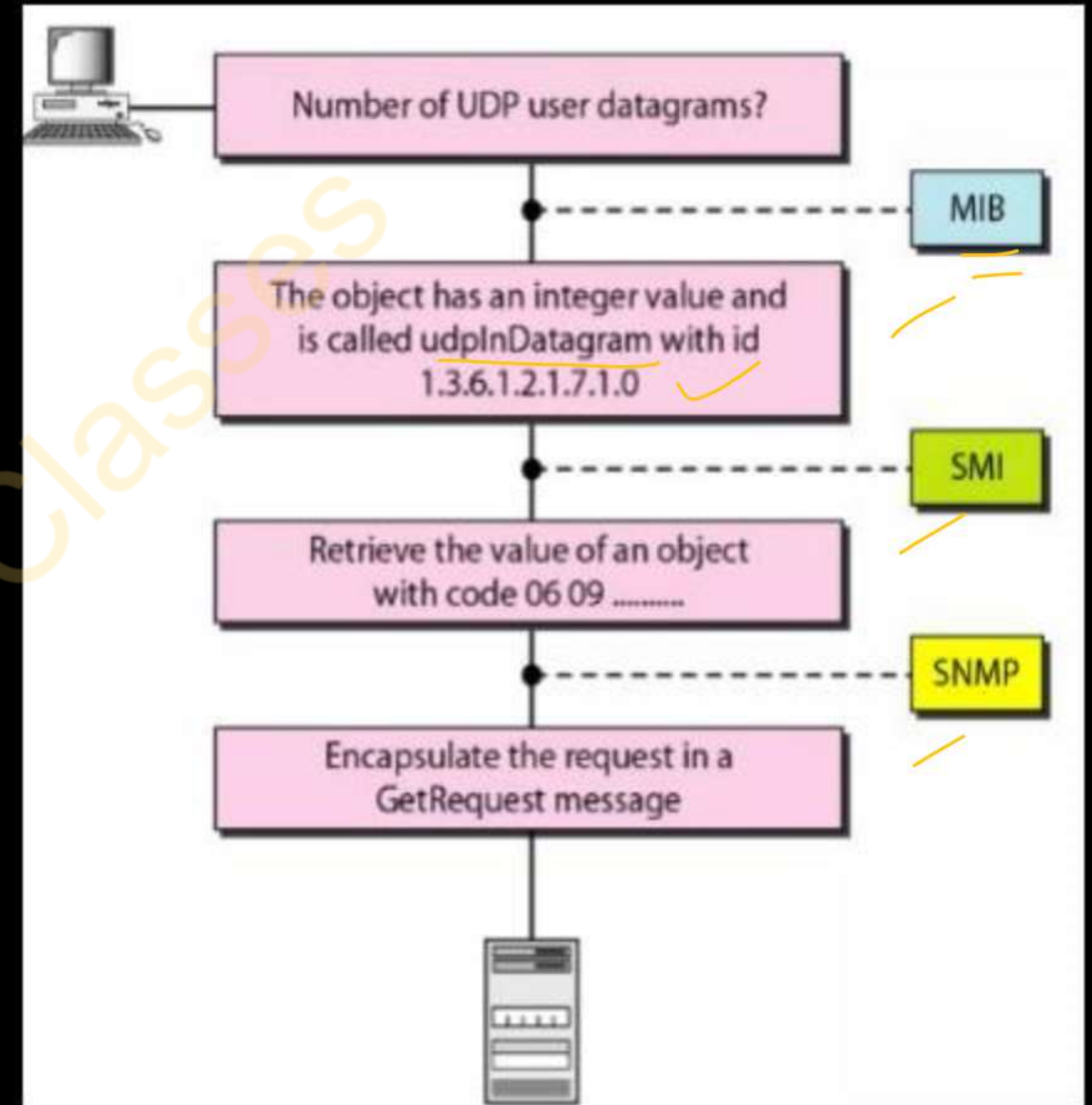
In network management, MIB names each object and defines type of the objects. type is already defined by SMI, so SNMP knows the range and size ✓

3. After declaration in programming, program needs to write statements to store values in the variables and change them if needed.

In network management, SNMP stores, changes, and interprets the values of objects already declared by MIB according to the rules defined by SMI

➤ Manager station (SNMP client) wants to send a message to an agent station (SNMP server) to find the number of UDP user datagrams received by the agent.

1. MIB is responsible for finding the object that holds number of UDP user datagrams received.
2. SMI, with help of another embedded protocol, is responsible for encoding the name of object.
3. SNMP is responsible for creating a GetRequest message, and encapsulating encoded message.



(AKTU)

List of common email gateways that are widely used for routing, filtering, and securing email traffic:

Cloud-Based Email Gateways: hosted by third-party providers and are easier to manage:

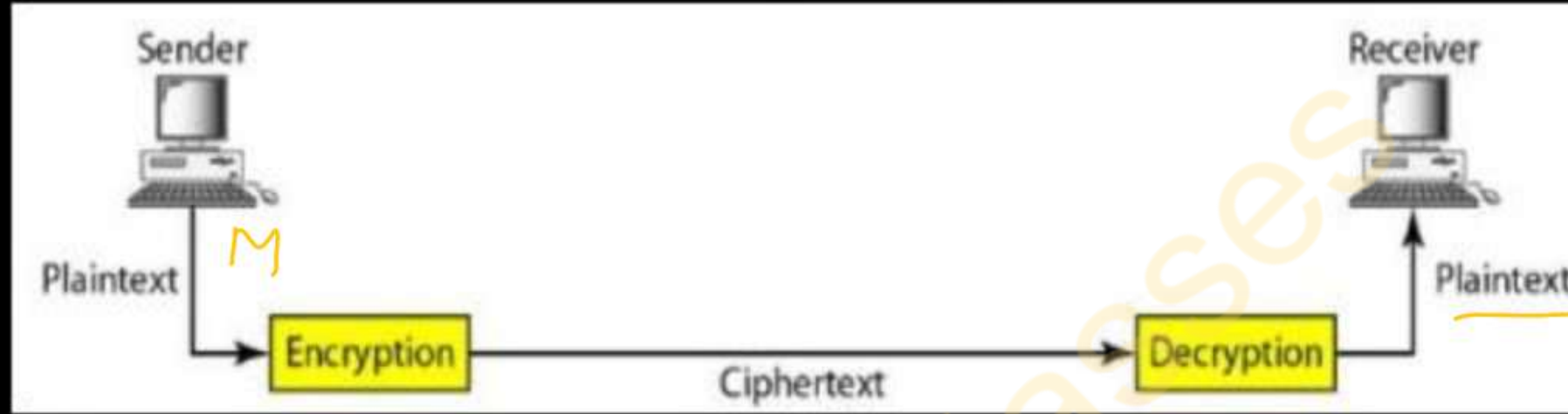
1. Microsoft Defender for Office 365
2. Google Workspace (Gmail) Security / Google Cloud Email Gateway
3. Proofpoint Essentials / Enterprise
4. Mimecast Email Security
5. Barracuda Email Security Gateway (Cloud version)
6. Cisco Secure Email (formerly Cisco Email Security / IronPort)
7. Trend Micro Hosted Email Security
8. Sophos Email
9. Zix Email Security
10. SpamTitan Cloud

On-Premises / Appliance-Based Email Gateways: These are deployed within your network:

1. Barracuda Email Security Gateway (Appliance)
2. Cisco Secure Email Appliance
3. Fortinet FortiMail
4. Symantec Messaging Gateway (Broadcom)
5. Trend Micro InterScan Messaging Security
6. McAfee Email Gateway (Legacy)
7. Axway SecureTransport (for email + file gateways)
8. Trustwave Secure Email Gateway

Cryptography

Cryptography is the science and art of transforming messages to make them secure and immune to attacks.



Plaintext and Ciphertext: The original message, before being transformed, is called plaintext.

- After the message is transformed, it is called ciphertext.
- An encryption algorithm at sender transforms the plaintext into ciphertext;
- A decryption algorithm at receiver transforms the ciphertext back into plaintext.

Key: A key is a number (or a set of numbers) that the cipher, as an algorithm, operates on.

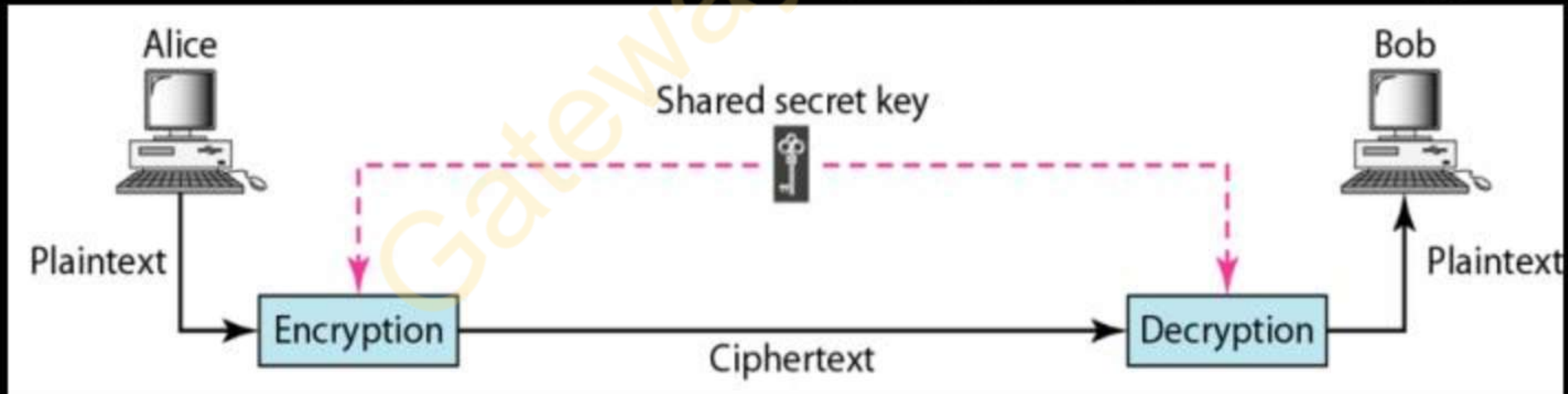
- To encrypt a message, we need encryption algo, encryption key, and plaintext to create ciphertext.
- To decrypt a message, we need decryption algo, decryption key, ciphertext to reveal plaintext.

Type of cryptography algorithms:

1. Symmetric key (also called secret-key) cryptography algorithms
2. Asymmetric (also called public-key) cryptography algorithms

Symmetric Key Cryptography:

- Same key is used by both parties. **The key is shared.**
- The sender uses this key and an encryption algorithm to encrypt data.
- Receiver uses the same key and the corresponding decryption algorithm to decrypt the data.

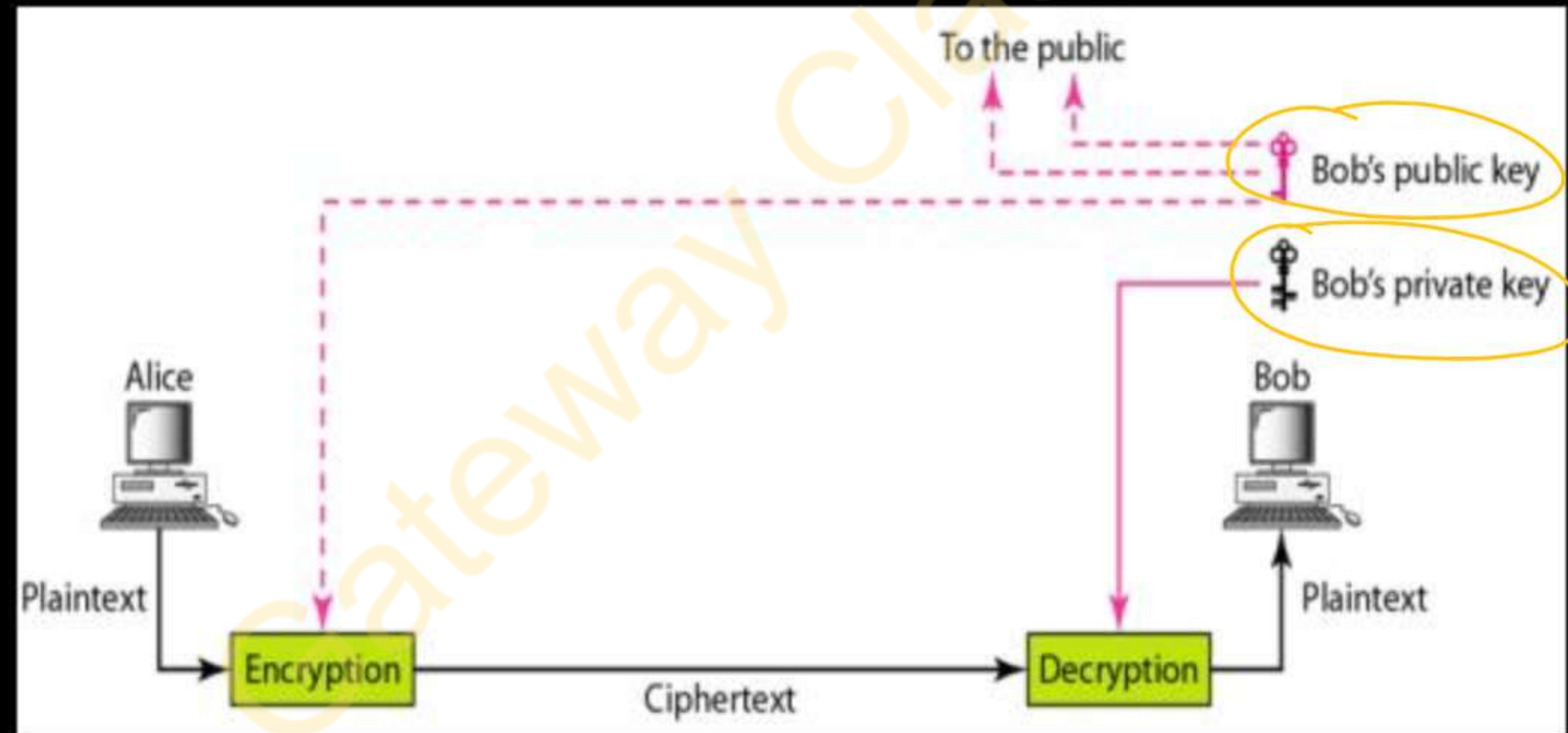


Asymmetric-Key Cryptography:

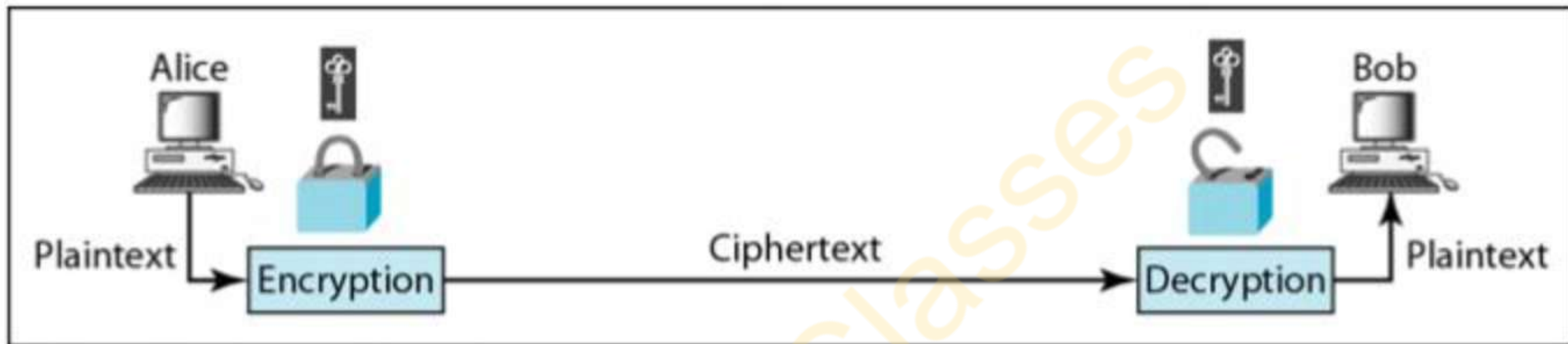
- There are two keys: a private key and a public key.
- The private key is kept by the receiver. The public key is announced to the public.

Example: Alice wants to send a message to Bob. Alice uses Bob's public key to encrypt the message.

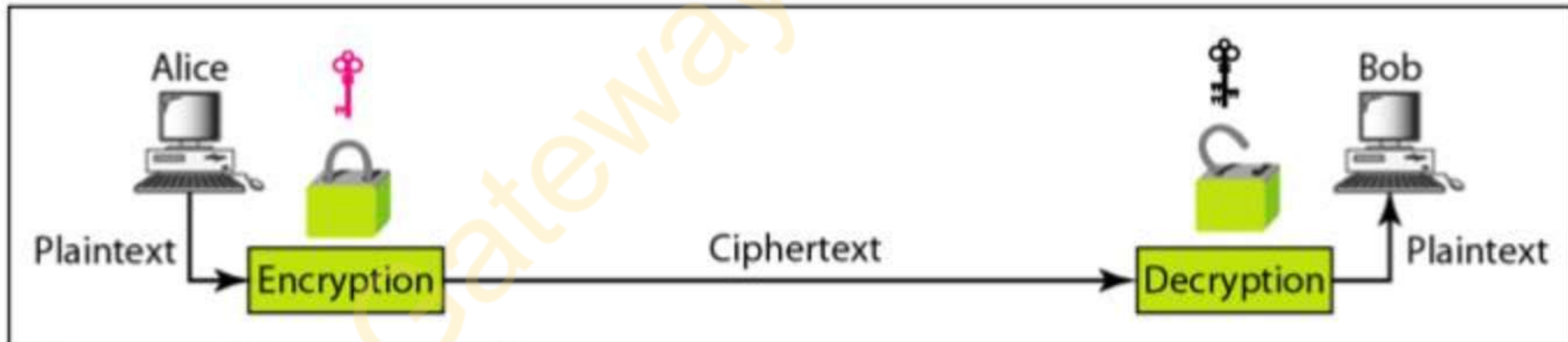
When the message is received by Bob, he uses his private key to decrypt the message.



Public key is available to the public; the private key is available only to an individual.

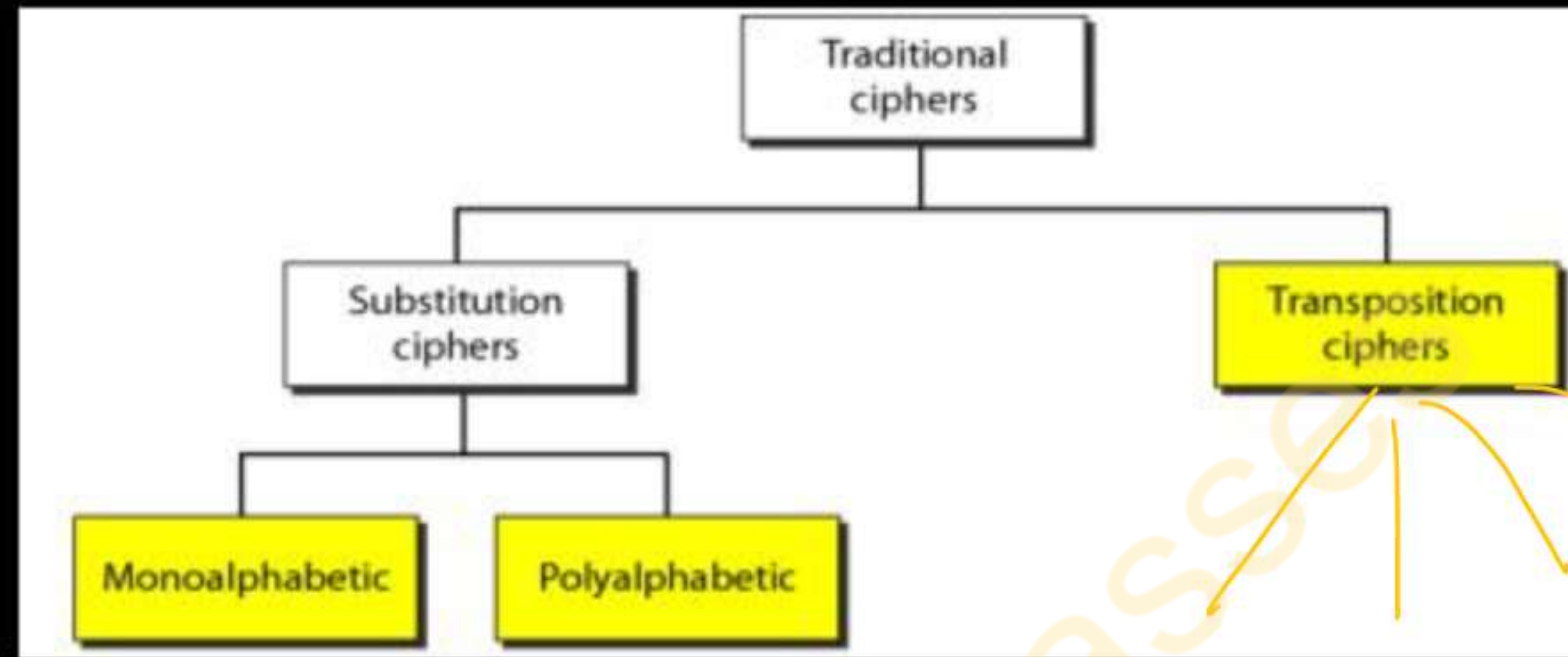


a. Symmetric-key cryptography



b. Asymmetric-key cryptography

Symmetric-key ciphers:



Substitution Cipher:

substitutes one symbol with another. e.g., we can replace character A with D, and character T with Z.
If the symbols are digits (0 to 9), we can replace 3 with 7, and 2 with 6.

Monoalphabetic cipher:

- A character (or a symbol) in plaintext is **always changed to same character** (or symbol) in the ciphertext regardless of its position in the text. If algorithm says that character A in the plaintext is changed to character D, every character A is changed to character D.
- Relationship between characters in plaintext and ciphertext is a one-to-one relationship.

Polyalphabetic cipher:

- Each occurrence of a character can have a different substitute.
- Relationship between a character in plaintext to a character in ciphertext is a one-to-many.
- For example, character A could be changed to D in the beginning of the text, but it could be changed to N at the middle.
- Key must tell us which of the many possible characters can be chosen for encryption.

For this, we need to divide the text into groups of characters and use a set of keys.

Example: we can divide the text "THISISANEASYTASK"

into groups of 3 characters and then apply the encryption using a set of 3 keys. We then repeat the procedure for the next 3 characters.

Q: Is the cipher monoalphabetic?

Plaintext: HELLO

Ciphertext: KHOOR

Solution

- The cipher is probably monoalphabetic because both occurrences of L's are encrypted as O's.

Q: Is the cipher monoalphabetic?

Plaintext: HELLO

Ciphertext: ABNZF

Solution

- The cipher is not monoalphabetic because each occurrence of L is encrypted by a different character.
- The first L is encrypted as N; the second as Z.

Transposition Ciphers:

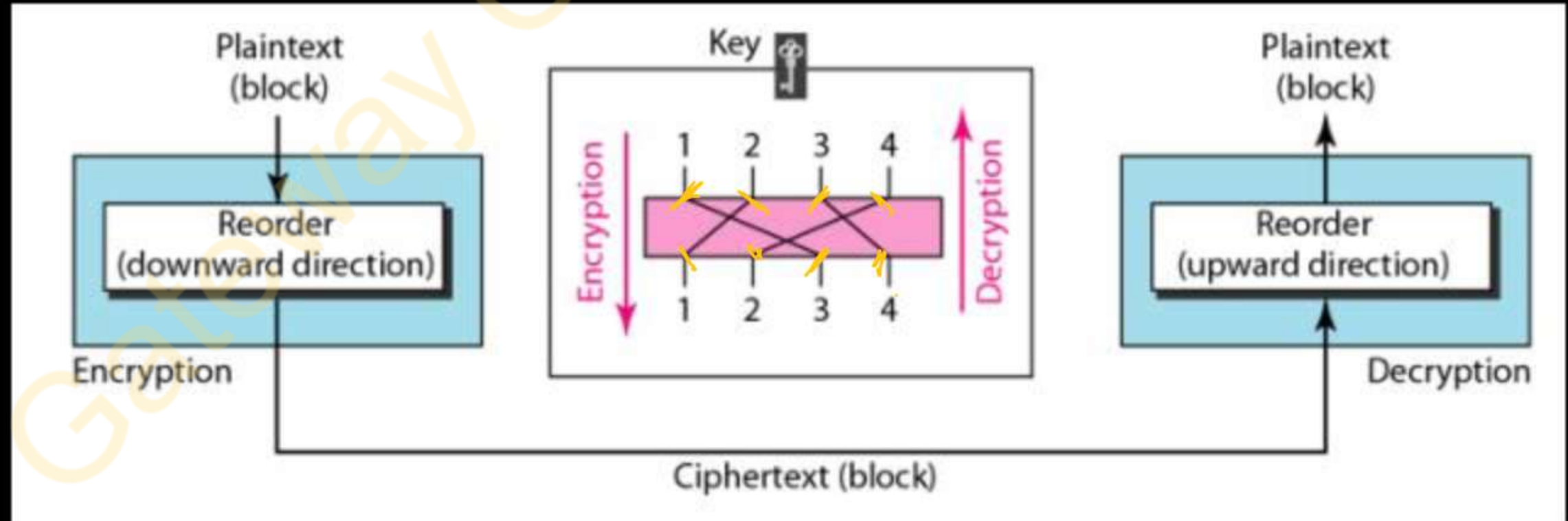
No substitution of characters; instead, their locations change.

A transposition cipher reorders the symbols in a block of symbols.

Key in a transposition cipher is a mapping between the position of the symbols in the plaintext and cipher text.

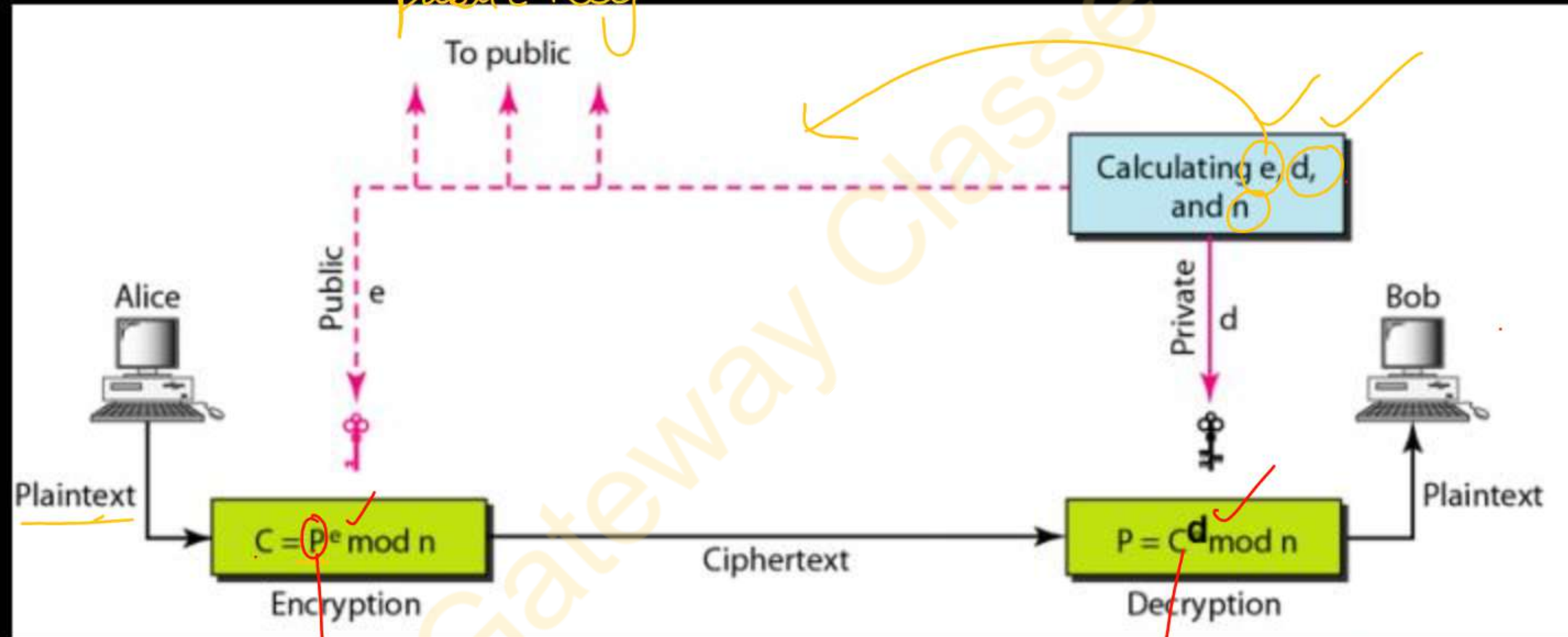
Example: the following shows the key using a block of four characters:

Plaintext: 2 4 1 3
Ciphertext: 1 2 3 4



RSA: (Imp)

The most common public key algorithm is RSA, named for its inventors Rivest, Shamir, and Adleman (RSA). It uses two numbers, e and d , as the public and private keys.



RSA Algorithm for selecting Keys

Bob use the following steps to select the private and public keys:

1. Bob chooses two very large prime numbers p and q .
2. Bob multiplies the above two primes to find n , the modulus for encryption and decryption.
$$n = p \times q$$
3. Bob calculates another number $\Phi(n) = (p - 1) \times (q - 1)$
4. Bob chooses a random integer e , such that $1 < e < \Phi(n)$, and $\gcd(e, \Phi(n)) = 1$, that is e should be co-prime with $\Phi(n)$.
$$d \equiv e^{-1} \pmod{\Phi(n)}$$
5. He then calculates d so that $(d * e) \equiv 1 \pmod{\Phi(n)}$ (d is modular multiplicative inverse of $e \pmod{\Phi(n)}$).
Can be calculated using Extended Euclidean Algorithm or Fermat's Little Theorem
We can have multiple values of d satisfying $(d * e) \equiv 1 \pmod{\Phi(n)}$ use any one of them.
6. Bob announces the **Public Key** = (n, e) and the keeps the secret **Private Key** = (n, d)

Encryption:

- Anyone who needs to send a message to Bob can use n and e .
- if Alice needs to send a message to Bob, she can change the message to an integer. This is the plaintext. She then calculates the ciphertext, using e and n .

$$C = p^e \pmod{n}$$

- Alice sends C , the ciphertext, to Bob.

Decryption:

- When Bob receives ciphertext, he uses his private key d to decrypt the message:

$$P = C^d \pmod{n}$$

Restriction:

For RSA to work, the value of P must be less than the value of n . If P is a large number, the plaintext needs to be divided into blocks to make P less than n .

RSA Example

Q-

① choose $p=3, q=11$

② calculate $n = p \times q$

$$n = 3 \times 11$$

$$\boxed{n = 33}$$

③ calculate $\phi(n) = (p-1) \times (q-1)$

$$\phi(n) = (3-1) \times (11-1)$$

$$= 2 \times 10$$

$$\boxed{\phi(n) = 20}$$

④ choose e such that $1 < e < \phi(n)$, $\gcd(e, \phi(n)) = 1$

prime no - $3, 7, 11, 13, 17, 19$

$$\boxed{1 < e < 20}$$

$$\boxed{e = 7}$$

RSA Example

(5)

$$d \cdot e \equiv 1 \pmod{\phi(n)}$$

$$\underline{d \cdot e \pmod{\phi(n)} = 1}$$

(I)

$$\underline{(d \cdot 7) \pmod{20} = 1}$$

$$\boxed{d=3}$$

$$\begin{array}{r} d \times 7 \\ \hline 21 \\ 42 \\ 63 \\ 84 \end{array}$$

(II)

$$d \cdot e \equiv 1 \pmod{\phi(n)}$$

$$d \cdot e = 1 + K \phi(n)$$

$$\boxed{d = \frac{1 + K \phi(n)}{e}}$$

constant

put $K=0/1/2/3 \dots$

$$\boxed{K=0} \Rightarrow d = \frac{1 + 0 \cdot 20}{7} = \frac{1}{7} = 0.1 \dots$$

$$\boxed{K=1} \Rightarrow d = \frac{1 + 1 \cdot 20}{7} = \frac{21}{7} = 3 \checkmark \checkmark \checkmark$$

(not considered as it is in fraction)

$$\boxed{d=3}$$

RSA Example

public key = (e, n) ✓

private key = (d, n)

Encryption →

$$C = P^e \bmod n$$

plaintext

Suppose $P = 2$

$$\begin{aligned} C &= 2^7 \bmod 33 \\ &= 128 \bmod 33 \end{aligned}$$

$C = 29$ ⇒ This ciphertext is sent from Alice to Bob.
over internet.

Decryption -

$$P = C^d \bmod n$$

$$P = 29^3 \bmod 33$$

$$P = 2$$

Advantages:

- Secure for data transfer.
- Two parties can exchange a secret key without actually sending the key over the network.
- Can be used for digital signatures, a sender can sign a message using their private key, and receiver can verify the signature using the sender's public key.
- Widely used in online banking, e-commerce, and secure communications.

Disadvantages:

- Slow processing speed with large amounts of data.
- Large key size required to be secure, so it requires more computational resources and storage space.
- Vulnerability to side-channel attacks, means an attacker can use information leaked through power consumption, electromagnetic radiation, and timing analysis to extract the private key.
- Not suitable for applications that require constant encryption and decryption of large amounts of data.

Data compression

Used to maximize the use of bandwidth across a network or to optimize disk space when saving data.

There are two general types of compression algorithms:

1. Lossless compression
2. Lossy compression

Lossless Compression:

- Compresses data in such a way that when data is decompressed there is no loss of data.
- Used to compress executable code, text files, and numeric data, because programs that process such file data cannot tolerate mistakes in the data.
- Does not compress file as much as lossy compression and may take more processing power.

Lossless Compression Algorithms:

1. Run length encoding:

- Replaces consecutive occurrences of a given symbol with only one copy of symbol along with a count of how many times that symbol occurs.
- For example, the string AAABBCDDDD would be encoded as 3A2B1C4D.
- Must be used only when lot of repetition in the otherwise it may increase the size of a file.

2. Differential pulse code modulation:

- First a reference symbol is placed. Then for each symbol in the data, we place the difference between that symbol and the reference symbol used.
- For example, using symbol A as reference symbol, the string AAABBC DDDD would be encoded as A0001123333.
A0001123333.

3. Dictionary based encoding:

- One is Lempel-Ziv (LZ) compression algorithm in which a dictionary (table) of variable length strings is built.
- This dictionary contains almost every string that is expected to occur in data.
- Let us say that the word “compression” has the index 4978 in one particular dictionary; To compress a body of text, each time the string “compression” appears, it would be replaced by 4978.

Lossy Compression:

- Does not promise that the data received is exactly the same as data send *i.e.* the data may be lost.
- This is because a lossy algorithm removes information that it cannot later restore.
- Used to compress still images, video and audio.
- Achieve much better compression ratios than the lossless algorithms.

Audio Compression: Used for speech or music.

Two types of techniques are used for audio compression:

Predictive encoding(used for speech):

- Differences between the samples are encoded instead of encoding all the sampled values.

Perceptual encoding:

- Used to create a CD-quality audio that requires a transmission bandwidth of 1.411 Mbps.
- MP3 (MPEG audio layer 3), a part of MPEG standard uses this perceptual encoding.
- Encode a signal in a way that it sounds the same to a human listener, even if it looks quite different on an oscilloscope.
- Some sounds can mask other sound. Like if you are broadcasting a live flute concert and all of a sudden someone starts striking a hammer on a metal sheet. You will not be able to hear the flute any more. Its sound has been masked by the hammer.

Masking can also be done on the basis of time.

For example: Even if the hammer is not striking on a metal sheet, the flute will be inaudible for a short period of time because the ears turn down its gain when they start and take a finite time to turn up again. Thus, a loud sound can numb our ears for a short time even after the sound has stopped. This effect is called temporal masking.

MP3: Uses frequency masking and temporal masking to compress audio signals.

- Divides the spectrum into several groups. Zero bits are allocated to the frequency ranges that are totally masked.
- A small number of bits are allocated to the frequency ranges that are partially masked.
- A larger number of bits are allocated to the frequency ranges that are not masked.
- Based on the range of frequencies in the original analog audio, MP3 produces three data rates:
96kbps, 128 kbps and 160 kbps

Networking Commands (AKTU 2021-22)

1. ipconfig command:

It displays information about the current network settings on the machine such as IPv4 and IPv6 address of all network interface cards (Ethernet adapters, WiFi adapters, virtual network adapters etc), MAC address, default gateway, subnet mask, DNS server, DHCP information etc.

Here are some different options of this command:

- **ipconfig /?** : Displays all available options.
- **ipconfig /all** : This will display info for ALL network connection adapters of the computer.
- **ipconfig /release** : Release current IPv4 addresses (assigned dynamically from a DHCP server).
- **ipconfig /release6** : Same as above but for the IPv6 address.
- **ipconfig /renew** : Used to request a new IP address from a DHCP server.
- **ipconfig /renew6** : Same as above but for the IPv6 address.

2. nslookup command:

- “nslookup” stands for “Name System Lookup” and is useful in obtaining Domain Name System (DNS) related information about a domain or about an IP address (reverse DNS lookup).

nslookup [domain name]: The most popular usage of this command is to find quickly the IP address of a specific domain name

Example: **nslookup www.networkstraining.com**

nslookup [IP Address]: Perform a reverse-DNS lookup and try to match the given IP address in the command with its corresponding domain name.

Example: **nslookup 8.8.8.8**

3. tracert command:

- “tracert” in Windows stands for “Trace Route”.
- In Linux, the same command is “traceroute”.
- It traces the path that a TCP/IP packet takes towards a destination target and shows some information (if available) of the routing nodes within this path.
- Like “ping” command, “tracert” sends also ICMP echo packets to the destination with varying Time-to-Live values.

tracert [domain or IP] : Traces the TCP/IP path to the specified destination target IP or domain.

Example:

tracert www.networkstraining.com

4. ping Command: useful for testing connectivity for both local computer from where you execute the command and also for a remote computer or server which you try to reach.

- If you try to “ping” your local default gateway IP address and you get replies back (ICMP echo replies), this means your local computer is properly connected to the network.
- If you “ping” a remote server on the Internet and you get replies back, it means that the remote server is properly connected to its network as well.
 - **ping [IP Address]** : By default, it will send 4 ICMP packets to the stated IP address.
 - Example: ping 8.8.8.8
 - **ping [hostname or domain]** : resolve name to IP address and then send ICMP packets to that IP.
 - Example: ping www.networkstraining.com
 - **ping [IP address] -t** : Send ping packets (ICMP echo requests) continuously to the target IP.
 - **ping -n 10 [IP address]** : This will send 10 ping packets (ICMP echo requests) to the target IP.

5. netstat command:

Network Statistics

It shows the established network TCP/IP connections of the local computer with remote hosts, open ports on the machine, the process ID (PID) of each connection etc.

netstat -ano : Displays all connections and listening ports (-a), addresses and ports in numerical form (-n) and also the process ID of each connection (-o).

6. **hostname Command**:

hostname command displays the hostname of the machine that is running it.

Thank
you